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RESEARCH ARTICLE



Model and solution approach to coordinate production-inventory strategies considering nonlinear price-sensitive demand: application to Canadian pulp and paper industry

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ABSTRACT

This study addresses a practical problem within a multi-level supply chain where a wide range of customers can be served through different strategies such as make-to-stock, make-to-order, or vendor-managed inventory. The customer demand is stochastic, and sensitive to pricing associated with different production-inventory strategies. We propose a two-stage stochastic mixed-integer non-linear programming model. In the first stage, decisions are made regarding the selection of production-inventory strategies and pricing to maximise the expected profit. The second stage involves decisions related to production, inventory, and distribution, which are used to evaluate the first-stage decisions under various scenarios with different levels of accuracy. To solve the model, a metaheuristic approach based on the Simulated Annealing algorithm is developed. To showcase the practical applicability of our model and solution approach, we use a real case study in a Canadian pulp and paper supply chain. The results revealed that both the production-inventory strategy assigned to customers and the sales price underwent changes across scenarios. Furthermore, we demonstrated that by implementing the SA algorithm, we could improve the initial profit by up to 1.43% through slight adjustments in the sales price and assigned strategies for customers.

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1. Introduction

Various production-inventory strategies, including Make-to-Stock (MTS), Make-to-Order (MTO), and Vendor-Managed Inventory (VMI), exist to handle customer demand. These strategies have a direct impact on both the level of service provided to customers and the efficiency of managing a production system. Tyan and Wee (2003) emphasised that enhancing customer service presents a significant challenge in the management of production and inventory systems. As noted by Huiskonen, Niemi, and Pirttilä (2003), in today's business landscape, many companies are striving to gain a competitive edge by not only addressing the needs of an expanding customer base with diverse logistics requirements, but also by focusing on the improvement of their logistics services. Peeters and van Ooijen (2020) argued that neither a pure MTS nor a MTO strategy is suitable as a production approach to address this situation. A growing number of companies are finding it necessary to embrace a mixed MTS/MTO production strategy, enabling them to integrate both MTS and MTO approaches within a single facility (Wang, Chen, and Wang 2007). However,

as highlighted by Kuthambalayan and Bera (2020a), the investigation of these systems within a single facility remains relatively scarce in the production literature and further research is required to delve deeper into the challenges and issues associated with these systems. Research in mixed MTS/MTO production has predominantly centred on product-wise production strategies. This involves tackling the question of devising production strategies for individual products, as exemplified in the work of Kuthambalayan and Bera (2020b). However, companies are increasingly employing mixed production strategies for the same products when serving various sales channels or customers, although this facet has received limited research attention, as noted by Zhang et al. (2013). In such customer-wise mixed production systems, a central concern revolves around determining the appropriate strategy to assign customers who may place different orders for various products. The customer-wise mixed production system presents a higher degree of complexity compared to the product-wise mixed production system, particularly when fulfilling various customer-specific product orders.

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Moreover, the VMI strategy is recognised as a well-established collaborative programme for the efficient management of a specific customer segment (Ghasemi, Lehoux, and Rönqvist 2023). However, there has been a notable gap in the existing literature regarding a thorough examination of how to identify potential customers suitable for inclusion in the VMI strategy. Taking into account a combination of VMI and non-VMI customers represents a significant extension to conventional VMI models. Weißhuhn and Hoberg (2021) highlighted the influence of altering the number of VMI customers within the system on both profit and service levels. In such a setting, a number of challenges must be considered. First, utilising the different production-inventory strategies to handle customer demand may lead to varied logistics services, and subsequently resulting in diverse prices for similar products. In competitive markets, price can impact customer demand, as customers' willingness to pay tends to decrease when significant disparities exist between the prices offered by companies and the market prices. Prices for identical products can fluctuate among various markets or customers. These price variations can mainly be attributed to differences between markets, as discussed by Adeinat et al. (2022), or they can be resulted from the provision of distinct services tailored to specific customer segments, as exemplified by the approach taken by Ali et al. (2019). In a customer-wise mixed production system, different pricing levels can be established for the customers served under MTS, MTO, or VMI strategies, based on different services provided to customers, including expedited delivery, reduced lead times, or personalised preferences. According to Duan and Ventura (2021), pricing decisions can have a substantial impact on the profitability of a firm.

Second, in practical business contexts, adjusting the price directly impacts customer demand, thereby affecting production, inventory, and distribution decisions (Adeinat et al. 2022). Deng and Yano (2006) demonstrated that coordinated decision-making in the context of pricing, production, and inventory offers advantages over decentralised solutions where these decisions are made independently. Similar results have been confirmed by Adeinat et al. (2022). This coordination enables businesses to effectively manage their resources, respond to market dynamics, and meet customer demand while maximising their financial performance. Furthermore, there is a direct relationship between production-inventory decisions and assigned strategy for production and inventory management, based on customer demand. Therefore, coordinating sales price with production-inventory strategy involves strategically aligning the pricing decisions with the selected strategy for managing production and inventory to optimise profitability.

Third, future customer demand is difficult to forecast and overestimating the demand can result in excessive inventories and increased costs associated with holding inventory, while underestimating the demand poses the risk of shortages and potential penalty costs (Weskamp et al. 2019). Demand represents a primary factor of uncertainty within production-inventory-distribution systems (Gupta and Maranas 2003) and pricing decisions exerts a notable influence on customer demand. The lack of transparent insight into demand has been widely recognised as a substantial impediment to supply chain management (Småros et al. 2003; Giri and Sarker 2019). Consequently, managing production and inventories based on inaccurate demand information can lead to suboptimal utilisation of capacity, insufficient product availability, and excessive stock levels. To effectively allocate resources and mitigate the risks associated with uncertain customers' demand, it becomes imperative to incorporate demand uncertainty into pricing and production-inventory decisions. Considering the price-dependent stochastic demand, businesses can tailor their pricing strategies to align with diverse production-inventory strategies, leading to improved customer satisfaction and loyalty.

Considering the main challenges addressed within customer-wise mixed production-inventory strategy systems, it is necessary to address two fundamental issues simultaneously to enhance the total profitability of such production systems. The first one entails determining the sales price for each production-inventory strategy. The second issue revolves around deciding which production-inventory strategy to assign to customers demand while accounting for the uncertain and price-sensitive nature of demand. Emphasising the interdependence between the assigned production-inventory strategy and the determined sales price, as well as their consequential impact on operational planning level activities encompassing production, inventory, and distribution, is of utmost importance.

The primary objective of this study is therefore to simultaneously examine pricing and allocation of production-inventory strategies to customers within a production system with mixed production-inventory strategies. We specifically focus on a multi-level production-inventory-distribution supply chain which is characterised by geographically dispersed customers and operates in the context of price-sensitive stochastic demand conditions. To this end, we develop a two-stage stochastic mixed-integer nonlinear programming (MINLP) model. This model serves to effectively coordinate sales pricing with production-inventory strategies, taking into account various demand scenarios. In the first stage, we make decisions regarding the selected strategy

and sales price for customer demand. In the second stage, these decisions are assessed in terms of production, inventory, and distribution, based on the expected profitability. In addition, due to the computationally intensive nature of simultaneously addressing these issues, a metaheuristic approach based on simulated annealing algorithm is utilised to solve the considered complex optimisation problem that arises from coordinating pricing and production-inventory decisions with stochastic and price-sensitive demand.

According to the recent survey articles by Hrabec, Hvattum, and Hoff (2022) and Ghasemi, Lehoux, and Rönnqvist (2023), there is a scarcity of studies in the integrated production, inventory, and distribution that investigate the real-world applications. These surveys underscored the necessity for researchers to undertake further investigations using real-world data. In this article, we apply the proposed model and approach on a large Canadian pulp and paper industry. The pulp and paper sector, as a part of the forest supply chain, is responsible for the production of a wide variety of papers and other cellulose-based fibre products which are used in various applications and sectors (Carlsson et al. 2009). Moreover, the production, inventory management, and distribution of these products necessitate the integration of a diverse set of activities. The integration of operations within the forest supply chain remains a significant challenge in the context of the broader forest products supply chain, as emphasised by Rönnqvist et al. (2015). Furthermore, the complexity of uncertainty within the forest products supply chain underscores its importance as a crucial factor to be taken into account in the decision-making process. As a whole, the pulp and paper industry serves as a good example for exploring mixed production strategies due to its essential need to provide a wide range of distinct and differentiated products for a broad spectrum of customers.

Our research demonstrates the impact of considering various demand scenarios on both the strategy assigned to customers and the sales price. The findings further illustrate that adjusting the sales price can result in an increase in overall profit. Last, the performance of the proposed solution approach validates the necessity of employing appropriate solution techniques to address problem instances according to real-world requirements. The outcomes of our study offer valuable insights for prospective users of our model, thereby simplifying its integration into industrial applications. The contribution of this article is threefold:

- Our primary contribution begins with the formulation of coordinating sales price with assigned production-inventory strategies to customers within a

customer-wise mixed MTS/MTO/VMI production systems in a multi-level, multi-product supply chain, an innovative concept that has yet to be explored in the current body of literature.

- Our second contribution involves the development of a two-stage stochastic MINLP model tailored to the problem at hand. This model takes into account a nonlinear price-sensitive demand function for customers within an uncertain demand market. In light of the computational complexity of solving this problem, we introduce a heuristic solution approach based on simulated annealing to solve the developed model.
- As a final contribution, in order to underscore the practical significance of our proposed model and solution approach, we present the computational results derived from an extensive case study conducted within a major Canadian pulp and paper industry.

It should be noted that the current version of the model is intentionally kept at a manageable complexity, and additional complexities, such as considering the routing problem for VMI customers – an acknowledged advantage of VMI implementation (Archetti and Grazia Speranza 2016) – have not been incorporated. The developed model and solution approach in this article could be extended to other industries sharing similar characteristics with the case company discussed in this article. In other words, within comparable supply chains, where production companies supply a diverse range of products to a broad customer base, there is an interest in categorising customers based on production-inventory strategies to enhance profitability and improve customer service. Therefore, the insights from this study could offer advantages to professionals and decision-makers in similar industries, allowing them to customise and align the findings with the specific production contexts within their respective sectors.

The remainder of this article is organised as follows. In Section 2, we provide a review of the relevant literature. In Section 3, we present the integrated pricing and production-inventory problem inspired from a real problem observed in practice. The structure of the model is outlined in Section 4. Section 5 introduces a metaheuristic algorithm designed to solve the developed model in the context of the real-world case study. In Section 6, we apply the proposed model and algorithm to a real case study in a Canadian pulp and paper industry, highlighting their efficacy and discussing the results. Section 7 delves into discussions regarding managerial insights. Finally, Section 8 encompasses concluding remarks and recommendations for future research directions.

2. Related literature

To enhance their logistics services, production industries focus on satisfying the demand of diverse customers by manufacturing a variety of products while concurrently improving product quality, cost-efficiency, and delivery speed (Günlal 2011). Aligned with this approach, numerous companies choose to adopt a production-inventory system that combines various production strategies, including MTS and MTO. However, in customer-wise production systems, alternative strategies like VMI have been explored to efficiently address the diverse product requirements of distinct customers. Within these systems, a multitude of tactical and operational decisions arise, encompassing the assignment of strategies to customers, determination of sales prices based on the assigned strategy at the tactical level, and the coordination of production, inventory, and distribution activities at the operational level, following the assigned strategy to the customer demand.

In the context of mixed production-inventory strategies, numerous studies have focused on coordinating operational planning decisions within product-wise mixed MTS/MTO production systems. In these studies, the central question revolves around determining which products should be produced using which specific strategy. Rajagopalan (2002) introduced a mathematical model that focused on a single-stage production facility, aiming to determine the optimal selection of products to be produced under MTS and MTO strategies. Kaminisky and Kaya (2009) investigated a multi-item two-stage supply chain scenario involving a single manufacturer and a single supplier. Their study revolved around the decision-making process to determine the appropriate MTS or MTO strategy for each item within the supply chain. In their study, Köber and Heinecke (2012) put forward a methodology that aimed to leverage the benefits of both MTS and MTO strategies. Their approach sought to determine the appropriate production strategy for volatile and seasonal demand scenarios, effectively combining the advantages of MTO and MTS. Kuthambalayan and Bera (2020b) devised a two-stage non-linear mathematical model that focused on the mixed MTS and MTO production strategies. Their model considered factors such as guaranteed delivery time and stochastic demand with the objective of maximising the expected profit.

Another research direction within mixed production-inventory strategies has concentrated on customer-wise production systems, where decisions are made regarding the assignment of production-inventory strategies to specific customers. As noted by Zhang et al. (2013), using mixed production strategies for the same products

has become increasingly common for companies to offer through different sales channels or customers. Huiskonen, Niemi, and Pirttilä (2003) formulated an analytical framework that examined the interplay between customer and product characteristics when determining production and inventory control strategies within a company. Choi and Cheol (2014) conducted a study on a manufacturer operating with both primary (MTO) and secondary (MTS) sales channels. In this context, the demand in the primary channel was considered stochastic, while the demand in the secondary channel exhibited a linear relationship with the sales price. The study explored the manufacturer's ability to generate additional demand through the secondary sales channel by offering reduced prices compared to the primary channel, with the objective of maximising the utilisation of excess capacity once the demand from the primary channel was fulfilled.

In comparison to production systems that focus on products and employ mixed MTS/MTO strategies, customer-wise production systems have not received as extensive investigation in the existing literature. The production literature also seems to lack considering VMI strategy along with mixed MTS/MTO strategies. However, few studies investigated mixed VMI and non-VMI strategies under the same model. The term 'non-VMI strategy' refers to a business approach that does not involve the use of VMI, without specifying the alternative strategy being employed for those customers. Smáros et al. (2003) employed discrete-event simulation techniques to assess the efficiency impact on manufacturers when considering VMI and non-VMI customers. Carlsson, Flisberg, and Rönnqvist (2014) examined the distribution and inventory challenge faced by a large pulp producer, where the producer assumes responsibility for a significant portion of their customers' inventories. The selection of customers to be served under a VMI strategy has been a topic explored in a limited number of studies. For instance, Borade and Sweeney (2015) conducted a case study focusing on the implementation of a partial VMI strategy. Their primary objective was to determine which retailers should be served by the VMI strategy to enhance customer service levels. However, they did not consider any alternative strategies for the rest of customers who were not included in the VMI approach.

While researchers have conducted comprehensive examinations of mixed MTS/MTO and VMI strategies individually, to the best of our current knowledge, the existing body of literature on customer-wise production systems does not seem to simultaneously integrate the MTS, MTO, and VMI strategies. In these systems, aside

from determining the suitable strategy for each customer, there exist several challenges that must be addressed to enhance the performance of such customer-wise production systems. One of the primary challenges involves making decisions regarding the selling prices based on the level of service provided under each strategy. This challenge is significant because price adjustments have ripple effects on product demand, which subsequently influences operational-level decisions such as production, distribution, and inventory management (Adeinat et al. 2022). The sales price differential between MTS and MTO customers and VMI and non-VMI customers have been considered in several studies. In a single-product, two-level supply chain, Ghose et al. (2005) conducted an analysis using game theory to examine how the pricing decisions made by retailers affect the manufacturer's choice between MTS and MTO strategies. Their analysis took into account that demand is influenced by pricing decisions. In the context of an MTS environment, Ali et al. (2019) classified customers into three distinct groups based on their priorities. The study assumed that high-priority customers were willing to pay a 10% premium above the market price in exchange for expedited order delivery. In another study, Weißhuhn and Hoberg (2021) investigated the concept of smart replenishment in VMI, focusing on a scenario involving a single manufacturer and multiple end customers with uncertain demand for a single product. Their research considered a predetermined mix of VMI and non-VMI customers with varying sales prices, service level expectations, and shortage costs.

Prior studies have regarded the sales price as a fixed parameter. However, the examination of sales price as a decision variable and the coordination of sales price with operational-level planning decisions have been explored in a limited number of studies, especially in the context of a single production-inventory strategy. Feng, D'Amours, and Beauregard (2008) formulated mathematical models based on mixed-integer programming (MIP) to integrate cross-functional planning in sales, production, distribution, and procurement for an oriented strand board manufacturing company. The models were specifically designed for a multi-site manufacturing system catering to MTO customers with deterministic seasonal demand patterns. In another study, Feng, D'Amours, and Beauregard (2010) proposed rolling horizon simulation models in a multi-site MTO-based manufacturing supply chain to analyse the performance of partially and fully integrated sales and operations planning. The findings of their study underscored the limitations of deterministic models in effectively supporting decision-making and assessing performance in dynamic real-world business settings. Taleizadeh, Noori-Daryan,

and Cárdenas-Barrón (2015) studied joint optimisation of pricing, inventory, and production decisions in a VMI system with deteriorating items. They assumed market demand for products is deterministic and price sensitive. It means that the customers' demand decreases by increasing the selling prices. Despite the inherent importance and complexity of customer-wise production strategy, none of the studies have explored pricing decisions in the context of mixed MTS/MTO/VMI strategies.

According to Huang, Leng, and Parlar (2013), pricing decisions have been identified as influential factors that can shape demand. Duan and Ventura (2021) emphasised the critical role of pricing in maximising profits within a supply chain, as it involves finding the right balance between product supply and demand. Making pricing decisions in supply chain management is a challenging task that can significantly impact the overall profitability of the entire supply chain.

Several studies have focused on examining the coordination of pricing and operational decisions, taking into account the impact of price-sensitive demand. One such study conducted by Adeinat and Ventura (2015) presented a mixed-integer nonlinear programming model. The objective of this model was to determine the optimal selling price and replenishment policy for a particular product in a supply chain featuring a single retailer and multiple potential suppliers, while considering the price sensitivity of demand.

Moreover, determining prices based on customer willingness to pay is a vital factor in determining the profitability of the supply chain, as highlighted by Adeinat et al. (2022). The demand function or price-response function is a critical factor when making pricing decisions as it enables companies to understand how customers react to price changes and estimate the quantity of products customers are willing to purchase at different price points. Phillips (2021) highlighted the commonly assumed distribution of customer willingness to pay, which resembles a bell-shaped curve with the peak centred around the prevailing market price. This pattern is observed in many price-response functions used in practical applications. One widely utilised price-response function is the logit function which exhibits a reverse S-shaped curve and is generated by a bell-shaped distribution of willingness to pay (Phillips 2021). Building upon this concept, Duan and Ventura (2021) proposed a two-stage piecewise-linear approximation technique as an extension of the study conducted by Adeinat and Ventura (2015). Their approach aimed to address the integrated problem of pricing, supplier selection, and inventory replenishment. They incorporated the logit function to model price-sensitive demand and conducted a comprehensive analysis of the suitability of the logit demand

function. In a similar vein, Adeinat et al. (2022) formulated a mixed-integer nonlinear programming model to maximise the total profit of a multi-stage supply chain comprising multiple retailers. The model considered a logit price-sensitive demand function for a single product.

Pricing decisions can become even more challenging in the presence of demand uncertainty, as accurately predicting customer purchasing behaviour at different price levels becomes difficult. In addition, the complexity of the supply chain further complicates demand forecasting. A stream of literature focused on the pricing problem with considering price-dependent stochastic demand. Lin and Wu (2014) introduced a two-stage stochastic programming model to address the integrated problem of pricing and operational planning, aiming to maximise the expected profit of a manufacturer under demand uncertainty. The first stage of the model was deterministic, where tactical decisions concerning pricing, inventory levels, and inventory locations are made. In addition, operational decisions related to component procurement and product manufacturing were determined. In the second stage, products were shipped to fulfil the realised demand. Liu, Shang, and Su (2016) examined the problem of channel coordination in a supply chain consisting of one manufacturer and one retailer, where the demand is stochastic and dependent on both price and effort. Jadidi, Jaber, and Zolfaghari (2017) formulated two mathematical programming models to investigate the problem of integrating pricing and inventory decisions in a single-period, single-product setting where product demand is subject to uncertainty and influenced by price variations. Jazinaninejad et al. (2019) proposed a methodology that coordinates production, pricing, and periodic review policy decisions within a supply chain involving a single manufacturer and a single buyer, considering price-sensitive stochastic demand. They observed that many existing studies on pricing strategies overlook the crucial aspect of demand uncertainty. Wang, Diabat, and Wu (2021) examined a supply chain configuration in which two competing suppliers offer substitutable products through a shared retailer. The retailer operates in an environment of price-dependent random demand and serves two types of customers: price-sensitive customers and loyal customers.

The summary of the main characteristics of this study and of the reviewed literature on production-inventory systems are shown in Table A1 (see the Appendix). Aligning with the review, we delineate the identified research gaps as follows:

- Some articles have explored customer-wise production systems, focusing on assigning production-inventory strategies to customers rather than products.

However, none of these articles have simultaneously addressed MTS, MTO, and VMI strategies.

- The coordination of pricing with production strategy has not been thoroughly examined in the context of mixed production-inventory strategy studies. We did not find any articles addressing this issue while considering differences in the strategies assigned to customers and their reactions to changes in prices.
- The consideration of demand sensitivity to price is a common aspect in studies that integrate selling prices with operational planning decisions. However, this has not been explored in the context of mixed production-inventory strategies, including MTS, MTO, and VMI.
- Additionally, only a few articles have investigated customer-specific mixed production-inventory strategy models based on real industrial cases.

This article contributes to the existing literature by delving into the joint pricing and production-inventory strategy assignment problem in a customer-wise production system, specifically considering geographically dispersed customers with stochastic and price sensitive demand. Significantly, our focus extends to the concurrent coordination of these decisions, a dimension not explored in previous research to the best of our knowledge. This exploration aims to offer valuable insights into effectively coordinating these crucial elements within supply chain management.

3. Problem description

The production-inventory-distribution problem considered in this study has been motivated from a real-world problem observed in a multi-product supply chain structure with multiple customers. Figure 1 shows an illustrative example of supply chain considered in this article. In this particular setup, P products are produced by U mills and are then dispatched to N customers who are geographically dispersed. Different production-inventory strategies, namely MTS, MTO, or VMI, can be employed to serve these customers. The supply chain being analysed also includes K cross docks and W external warehouses. In addition, the transportation of products can be carried out using various modes of transportation.

Accurately forecasting customer demand becomes a challenging task for the company due to the wide range of products they produce. Such uncertainty in demand can lead to either overestimation or underestimation, resulting in increased inventory levels or penalties for shortages (Weskamp et al. 2019). To address this demand uncertainty, our study focuses on considering stochastic demand for customers. In addition, the actual demand

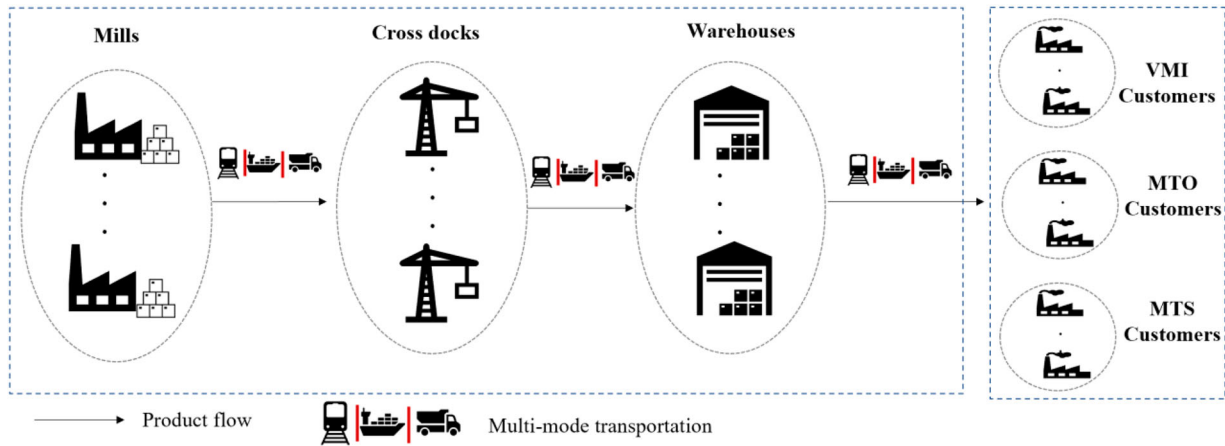


Figure 1. The schematic representation of the supply chain structure examined in this study.

from customers can vary based on the prices offered by the company for different production-inventory strategies, indicating that customers have their own price-sensitive demand functions. Phillips (2021) highlighted the commonly observed price-response functions, such as linear, power, exponential, constant elasticity, logit, and probit, each carrying its own set of advantages and disadvantages. The author stated that while the linear model is convenient and easily manageable, it falls short as a realistic global representation of price response. Phillips (2021) also pointed out that in constant-elasticity price-response functions, demand does not diminish to zero at any price, no matter how high, and as the price approaches zero, demand theoretically tends toward infinity. This suggests a potentially infinite demand. However, due to this characteristic, the constant-elasticity price-response function does not align with any clearly defined distribution of willingness to pay. In real-world scenarios, it is expected that customers' willingness to pay typically follows a bell-shaped distribution, peaking around the market price. This distribution generates a reverse S-shaped price-response function. Phillips (2021) suggested that this price-response function is applicable to a wide range of markets in terms of observed price response. In our study, we specifically consider the logit price-response function, which is the most widely used reverse S-shaped function (as shown in Equation (1)).

$$\text{demand}(\text{price}) = \frac{Ae^{-(a+b*\text{price})}}{1 + e^{-(a+b*\text{price})}} \quad (1)$$

In Equation (1), the market size is denoted as $A > 0$, while $b \geq 0$ represents the price sensitivity parameter and a represents the width of the demand curve. Higher values of b indicate a greater level of price sensitivity within the market. The logit price-response function has been extensively applied in various real-world scenarios, as

evidenced in the literature (e.g. Phillips 2021; Adeinat et al. 2022; Duan and Ventura 2021).

At the operational planning level, a company that possesses mills and external warehouses takes charge of managing the product flows and is responsible for making decisions regarding production, inventory, and distribution planning. This planning problem involves multiple decision-making aspects which can be summarised as follows. First, at the tactical planning level, the company needs to determine which production-inventory strategy (i.e. VMI, MTO, and MTS) should be employed for serving each customer. It is important to note that backorders or lost sales are not allowed for customers served through the VMI approach. However, for MTO customers, any unfulfilled demands are fully backordered and must be satisfied by the end of the planning horizon while ensuring a minimum service level. Conversely, in the case of MTS customers, any unmet demand is considered as lost sales, and there may be delays in delivering the products to these customers. Second, pricing decisions are made in relation to the selected production-inventory strategy. These decisions entail choosing the appropriate production-inventory strategy and setting the selling price for each product and customer. Last, in order to assess the effectiveness of these decisions, a production-inventory-distribution problem is solved over a planning horizon of T .

In the context of VMI, the inventory control policy of VMI customers emerges as a critical concern. Scholars have delved into various policies, such as the continuous review (r,q) policy, which determines the optimal reorder point for buyers (see for example, Bichescu and Fry 2009; Guan and Zhao 2010) or delivery products to the buyers based on their replenishment cycle (see for instance, Taleizadeh, Noori-Daryan, and Cárdenas-Barrón 2015). Maximum replenishment quantity is another inventory policy that represents an inventory strategy that offers

greater flexibility for the company to manage inventory levels with VMI customers. Under this approach, a predefined storage capacity is set for VMI customers, and exceeding this limit results in penalties for the company (see for example, Darwish and Odah 2010; Hariga, Gumus, and Daghfous 2014; Chakraborty, Chatterjee, and Mateen 2015). In addition, to establish an upper limit for inventory levels at VMI customers, Fry, Kapuscinski, and Olsen (2001) and Weißhuhn and Hoberg (2021) included a minimum inventory level in their analyses to ensure the protection of VMI customers against the risk of understocking. In our study, we assume that the production company is responsible for maintaining the inventory level at VMI customers within a specified upper and lower limits.

This study focuses on the development of a two-stage stochastic MINLP model. The objective is to effectively coordinate the selection of production-inventory strategies with pricing decisions in the presence of stochastic and price-sensitive demand, with the aim of maximising the expected profit. In this framework, the strategy selection and pricing decisions are regarded as first-stage decisions, while the production, inventory, and distribution decisions, which are utilised to evaluate the outcomes of the first-stage decisions, are considered as second-stage decisions (refer to Figure 2).

The validity of the proposed model in this study relies on the following assumptions:

- Each product can be produced using either an MTS, MTO, or VMI strategy. As a result, there is no longer lead time for MTO customers.

- The unit production times for all products are equal and are specified as a single period.
- The timely satisfaction of VMI customer demand is essential, and backorders are considered for MTO customers as long as the minimum service level is met. Unmet demand from MTS customers is treated as lost sales.
- Once a strategy is assigned to a customer, it applies to all products under their demand.
- The inventory level at the location of VMI customers should not exceed predetermined upper and lower bounds.
- At the end of each period, there is no inventory left stored at the cross docks.
- Demand is uncertain and sensitive to the price.
- Time horizon is finite.

4. Model development

In this section, we present the formulation of the two-stage stochastic MINLP model (referred to as $M0$). The primary objective of this model is to assist decision makers in determining the most suitable production-inventory strategy and pricing decisions for customers. The aim is to effectively manage production, inventory, and distribution activities within the supply chain. These decisions must be made based on available information, specifically the predicted customer demand, prior to the beginning of the planning period. It is crucial to note that the remaining logistics activities, such as production, inventory, and distribution planning, will be executed

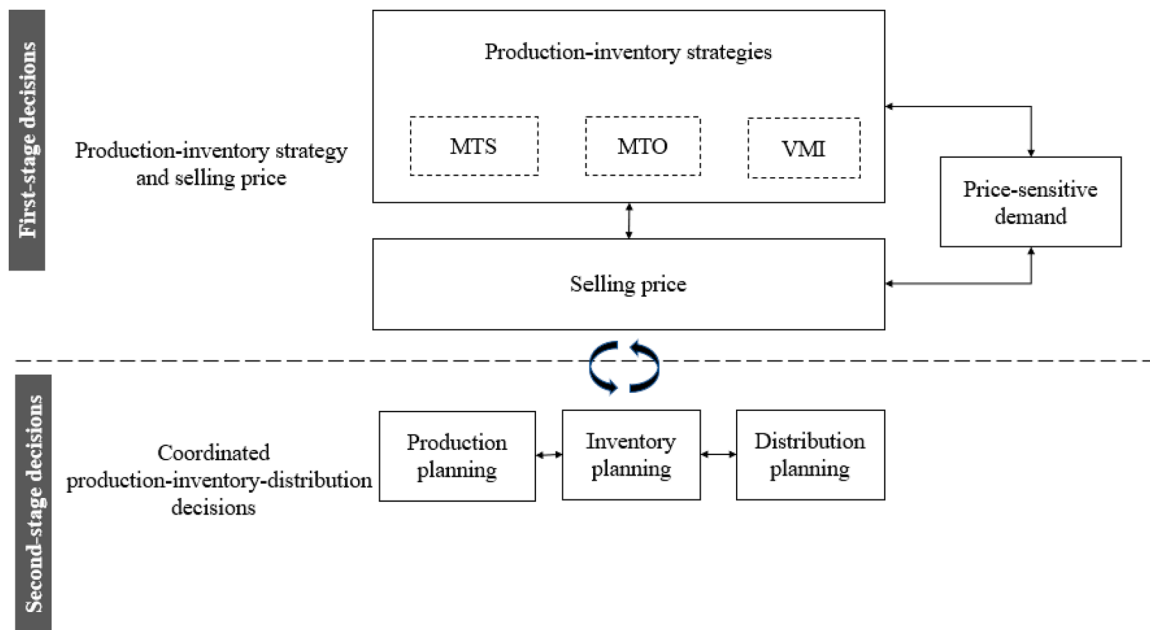


Figure 2. Decisions involved in the planning algorithm.

using the actual demand and are based on the assigned production-inventory strategy, as well as the agreed selling price between the company and its customers.

Within the model, the assignment of production-inventory strategy (MTS, MTO, and VMI) to customer i is represented by decision variables y_i^{mts} , y_i^{mto} , and y_i^{vmi} , respectively. In addition, the selling price decisions for customers i and product p is denoted by $e_{p,i}$. To evaluate the profitability of different production-inventory strategies for customers (i.e. first-stage decisions), a production-inventory-distribution problem is solved. For this purpose, second-stage decision variables are utilised, which include quantities produced, stored, backordered, and shipped. These variables are defined in various scenarios.

The mathematical formulation of the production-inventory-distribution problem under consideration involves various index sets, parameters, and decision variables, which are presented as follows:

Index Sets

N	Index set of customers
P	Index set of products
U	Index set of mills
K	Index set of cross-docks
W	Index set of warehouses
T	Index set of time periods
T^{mts}	Index set of time periods for MTS customers
V	Index set of transportation modes
Ω	Index set of scenarios
R^1	Index set of all mills and warehouses ($U \cup W$)
R^2	Index set of all mills and cross docks ($U \cup K$)
R^3	Index set of all cross docks and warehouses ($K \cup W$)
R^4	Index set of all mills, cross docks, and warehouses ($U \cup K \cup W$)
R^5	Index set of all mills, warehouses, and customers ($U \cup W \cup N$)
R^6	Index set of all cross docks, warehouses, and customers ($K \cup W \cup N$)

Deterministic parameters

g_i^{inv}	Unit inventory holding cost at node $i \in R^5$ per time unit
$g_{i,j,v}^{trs}$	Unit transportation cost between node $i \in R^4$ and node $j \in R^6$ using transportation mode $v \in V$
$g_{p,i}^{prd}$	Unit production cost of product $p \in P$ at production unit $i \in U$
w_p^{mto}	Unit backorder cost of product $p \in P$ for customers served by MTO strategy per time unit

o	Unit order management cost for customers served by MTO strategy
$s_{p,i}$	Setup cost for product $p \in P$ at production unit $i \in U$
c_i^{str}	Storage capacity at node $i \in R^5$ at each time unit
$c_{p,i}^{cap}$	Production capacity of product $p \in P$ at production unit $i \in U$ per time unit
c_i^{tcap}	Total production capacity at production unit $i \in U$ per time unit
$c_{i,j,v}^{trs}$	Maximum transportation capacity using transportation mode v between nodes i and j per time unit
c_i^{lod}	Loading capacity at production unit $i \in U$ per time unit
l_v	Shipment lead time in number of time periods using transportation mode $v \in V$
$a_{i,j,v}$	Transportation time from node $i \in R^4$ and node $j \in R^6$ using transportation mode $v \in V$
$b_{p,i}^{min}$	Minimum inventory level of product $p \in P$ at location for customer $i \in N$ served by VMI strategy
$b_{p,i}^{max}$	Maximum inventory level of product $p \in P$ at location for customer $i \in N$ served by VMI strategy
e^{mto}	Minimum service level expectation for customers served by MTO strategy
a_i	Width of the demand curve of customer $i \in N$
b_i	The price sensitivity of customer $i \in N$

Stochastic parameters

$d_{p,i,t,\omega}$	Stochastic market size of product $p \in P$ for customer $i \in N$ in period $t \in T$ in scenario $\omega \in \Omega$
π_ω	Probability of scenario $\omega \in \Omega$, subject to $\sum_{\omega \in \Omega} \pi_\omega = 1; 0 \leq \pi_\omega \leq 1$

First stage decision variables

$e_{p,i}$	Sales price for product $p \in P$ for customer $i \in N$
y_i^{vmi}	Binary variables equal to 1 if customer $i \in N$ is served under VMI policy, 0 otherwise
y_i^{mto}	Binary variables equal to 1 if customer $i \in N$ is served under MTO policy, 0 otherwise
y_i^{mts}	Binary variables equal to 1 if customer $i \in N$ is served under MTS policy, 0 otherwise

Second stage decision variables

$d_{p,i,t,\omega}^{price}$	Price-sensitive stochastic demand of product $p \in P$ for customer $i \in N$ in period $t \in T$ in scenario $\omega \in \Omega$
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$r_{p,i,t,\omega}$	Binary variables equal to 1 if demand of product $p \in P$ of customer $i \in N$ served under MTS strategy at period $t \in T$ in scenario $\omega \in \Omega$ is satisfied, 0 otherwise
$q_{p,i,t,\omega}$	Quantity of products $p \in P$ to be produced at production unit $i \in U$ at the beginning of period $t \in T$ in scenario $\omega \in \Omega$
$x_{p,i,j,v,t,t',\omega}$	Quantity of products $p \in P$ to be transported from node $i \in R^4$ at the end of period $t \in T$ to node $j \in R^3$ at the beginning of period $t' \in T$ using transportation mode $v \in V$ in scenario $\omega \in \Omega$
$x_{p,i,j,v,t,t',\omega}^{vmi}$	Quantity of product p to be transported from node $i \in R^4$ at the end of period $t \in T$ to customer $j \in N$ served by VMI strategy at the beginning of period $t' \in T$ using transportation mode $v \in V$ in scenario $\omega \in \Omega$
$x_{p,i,j,v,t,t',\omega}^{mto}$	Quantity of product $p \in P$ to be transported from node $i \in R^4$ at the end of period $t \in T$ to customer $j \in N$ served by MTO strategy at the beginning of period $t' \in T$ using transportation mode $v \in V$ in scenario $\omega \in \Omega$
$x_{p,i,j,v,t,t',\omega}^{mts}$	Quantity of product $p \in P$ to be transported from node $i \in R^4$ at the end of period $t \in T$ to customer $j \in N$ served by MTS strategy at the beginning of period $t' \in T$ using transportation mode $v \in V$ in scenario $\omega \in \Omega$
$h_{p,i,t,\omega}$	Inventory level of product $p \in P$ at node $i \in R^5$ at the end of period $t \in T$ in scenario $\omega \in \Omega$
$h_{p,i,0,\omega}$	Initial inventory level of product $p \in P$ at node $i \in R^5$ in scenario $\omega \in \Omega$
$s_{p,i,t}^{mto}$	Backlogged order quantity of product $p \in P$ for customer $i \in N$ served by MTO strategy in period t in scenario $\omega \in \Omega$
$u_{i,\omega}^{mto}$	Service level for customer $i \in N$ served by MTO strategy in scenario $\omega \in \Omega$

The formulation of the model M_0 is presented as follows:

$$\text{Maximise } \sum_{\omega \in \Omega} \pi_{\omega} \vartheta_{\omega} \quad (2)$$

Subject to

$$\vartheta_{\omega} = \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in N} \sum_{v \in V} \sum_{\substack{t, t' \in T \\ t' = t - a_{i,j,v} - l_v}} e_{p,j} \times x_{p,i,j,v,t,t',\omega}^{vmi}$$

$$\begin{aligned} & + \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in N} \sum_{v \in V} \sum_{\substack{t, t' \in T \\ t' = t - a_{i,j,v} - l_v}} e_{p,j} \\ & \times x_{p,i,j,v,t,t',\omega}^{mto} \\ & + \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in N} \sum_{v \in V} \sum_{\substack{t' \in T \\ t' = t - a_{i,j,v} - l_v}} \sum_{t \in T^{mts}} e_{p,j} \\ & \times x_{p,i,j,v,t',\omega}^{mts} \\ & - \sum_{p \in P} \sum_{i \in U} \sum_{t \in T} g_{p,i}^{prd} \times q_{p,i,t,\omega} - \sum_{i \in R^5} \sum_{p \in P} \sum_{t \in T} g_i^{inv} \\ & \times h_{p,i,t,\omega} - \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in R^3} \sum_{v \in V} \sum_{t, t' \in T} g_{i,j,v}^{trs} \\ & \times x_{p,i,j,v,t',\omega} \\ & - \sum_{p \in P} \sum_{i \in N} \sum_{t \in T} w_p^{mto} \times s_{p,i,t,\omega}^{mto} \\ & - \sum_{p \in P} \sum_{l \in R^4} \sum_{j \in N} \sum_{t, t' \in T} o \times x_{p,l,j,v,t',\omega}^{mto} \\ & - \sum_{p \in P} \sum_{i \in U} \sum_{t \in T} s_{p,i} \times q_{p,i,t,\omega} \\ & - \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in N} \sum_{v \in V} \sum_{t, t' \in T} g_{i,j,v}^{trs} \times x_{p,i,j,v,t',\omega}^{vmi} \\ & - \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in N} \sum_{v \in V} \sum_{t, t' \in T} g_{i,j,v}^{trs} \times x_{p,i,j,v,t',\omega}^{mto} \\ & - \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in N} \sum_{v \in V} \sum_{t' \in T} \sum_{t \in T^{mts}} g_{i,j,v}^{trs} \times x_{p,i,j,v,t',\omega}^{mts} \\ & \forall \omega \in \Omega \end{aligned} \quad (3)$$

$$\begin{aligned} & - \sum_{j \in N} \sum_{v \in V} \sum_{\substack{t' \in T^{mts} \\ t' = t + a_{i,j,v} + l_v}} x_{p,i,j,v,t',\omega}^{mts} \\ & \forall i \in U, \forall p \in P, \forall t \in T, \forall \omega \in \Omega \end{aligned} \quad (4)$$

$$\begin{aligned} & \sum_{j \in R^1} \sum_{v \in V} \sum_{\substack{t' \in T \\ t' = t - a_{i,j,v} - l_v}} x_{p,j,i,v,t',\omega} \\ & = \sum_{j \in W} \sum_{v \in V} \sum_{\substack{t' \in T \\ t' = t + a_{i,j,v} + l_v}} x_{p,i,j,v,t',\omega} \\ & + \sum_{j \in N} \sum_{v \in V} \sum_{\substack{t' \in T \\ t' = t + a_{i,j,v} + l_v}} (x_{p,i,j,v,t,t',\omega}^{vmi} + x_{p,i,j,v,t,t',\omega}^{mto}) \\ & + \sum_{j \in N} \sum_{v \in V} \sum_{\substack{t' \in T^{mts} \\ t' = t + a_{i,j,v} + l_v}} x_{p,i,j,v,t,t',\omega}^{mts} \\ & \forall i \in K, \forall p \in P, \forall t \in T, \forall \omega \in \Omega \end{aligned} \quad (5)$$

$$\begin{aligned}
h_{p,i,t,\omega} &= h_{p,i,t-1,\omega} + \sum_{j \in R^2} \sum_{v \in V} \sum_{\substack{t' \in T \\ t' = t - a_{ij,v} - l_v}} x_{p,j,i,v,t',\omega} \\
&\quad - \sum_{j \in K} \sum_{v \in V} \sum_{\substack{t' \in T \\ t' = t + a_{ij,v} + l_v}} x_{p,i,j,v,t',\omega} \\
&\quad - \sum_{j \in N} \sum_{v \in V} \sum_{\substack{t' \in T \\ t' = t + a_{ij,v} + l_v}} (x_{p,i,j,v,t',\omega}^{vmi} + x_{p,i,j,v,t',\omega}^{mto}) \\
&\quad - \sum_{j \in N} \sum_{v \in V} \sum_{\substack{t' \in T^{mts} \\ t' = t + a_{ij,v} + l_v}} x_{p,i,j,v,t',\omega}^{mts} \\
&\quad \forall i \in W, \forall p \in P, \forall t \in T, \forall \omega \in \Omega
\end{aligned} \tag{6}$$

$$\begin{aligned}
h_{p,i,0,\omega} &\leq \sum_{j \in N} \sum_{t \in T} d_{p,j,t,\omega} / (|N| * T) \\
&\quad \forall i \in R^1, \forall p \in P, \forall \omega \in \Omega
\end{aligned} \tag{7}$$

$$h_{p,i,T,\omega} \geq h_{p,i,0,\omega} \quad \forall i \in R^1, \forall p \in P, \forall \omega \in \Omega \tag{8}$$

$$\begin{aligned}
h_{p,i,t,\omega} &= h_{p,i,t-1,\omega} + \sum_{j \in R^4} \sum_{v \in V} \sum_{\substack{t' \in T \\ t' = t - a_{j,i,v} - l_v}} \\
&\quad \times x_{p,j,i,v,t',\omega}^{vmi} - d_{p,i,t,\omega}^{price} \times y_i^{vmi} \\
&\quad \forall i \in N, \forall p \in P, \forall t \in T, \forall \omega \in \Omega
\end{aligned} \tag{9}$$

$$\begin{aligned}
&\sum_{j \in R^4} \sum_{v \in V} \sum_{\substack{t' \in T \\ t' = t - a_{j,i,v} - l_v}} x_{p,j,i,v,t',\omega}^{mto} - s_{p,i,t-1,\omega}^{mto} \\
&= d_{p,i,t,\omega}^{price} \times y_i^{mto} - s_{p,i,t,\omega}^{mto} \\
&\quad \forall i \in N, \forall p \in P, \forall t \in T, \forall \omega \in \Omega
\end{aligned} \tag{10}$$

$$\begin{aligned}
&\sum_{j \in R^4} \sum_{v \in V} \sum_{\substack{t' \in T^{mts} \\ t' = t + a_{j,i,v} + l_v}} x_{p,j,i,v,t',\omega}^{mts} = d_{p,i,t,\omega}^{price} \times r_{p,i,t,\omega} \\
&\quad \forall i \in N, \forall p \in P, \forall t \in T, \forall \omega \in \Omega
\end{aligned} \tag{11}$$

$$\begin{aligned}
d_{p,i,t,\omega}^{price} &= d_{p,i,t,\omega} \frac{e^{-(a_i + b_i e_{p,i})}}{1 + e^{-(a_i + b_i e_{p,i})}} \\
&\quad \forall i \in N, \forall p \in P, \forall t \in T, \forall \omega \in \Omega
\end{aligned} \tag{12}$$

$$r_{p,i,t,\omega} \leq y_i^{mts} \quad \forall i \in N, \forall p \in P, \forall t \in T, \forall \omega \in \Omega \tag{13}$$

$$y_i^{vmi} + y_i^{mto} + y_i^{mts} = 1 \quad \forall i \in N, \forall \omega \in \Omega \tag{14}$$

$$s_{p,i,T,\omega}^{mto} = 0 \quad \forall i \in N, \forall p \in P, \forall \omega \in \Omega \tag{15}$$

$$q_{p,i,t,\omega} \leq c_{p,i}^{cap} \quad \forall i \in U, \forall p \in P, \forall t \in T, \forall \omega \in \Omega \tag{16}$$

$$\sum_{p \in P} q_{p,i,t,\omega} \leq c_i^{tcap} \quad \forall i \in U, \forall t \in T, \forall \omega \in \Omega \tag{17}$$

$$\sum_{p \in P} h_{p,i,t,\omega} \leq c_i^{str} \quad \forall i \in R^5, \forall t \in T, \forall \omega \in \Omega \tag{18}$$

$$\begin{aligned}
\sum_{p \in P} \sum_{\substack{t' \in T \\ t' = t + a_{ij,v} + l_v}} x_{p,i,j,v,t',\omega} &\leq c_{i,j,v}^{trs} \quad \forall i \in R^4, \forall j \in R^3, \\
&\quad \forall v \in V, \forall t \in T, \forall \omega \in \Omega
\end{aligned} \tag{19}$$

$$\begin{aligned}
&\sum_{p \in P} \sum_{\substack{t' \in T \\ t' = t + a_{ij,v} + l_v}} (x_{p,i,j,v,t',\omega}^{vmi} + x_{p,i,j,v,t',\omega}^{mto} + x_{p,i,j,v,t',\omega}^{mts}) \\
&\leq c_{i,j,v}^{trs} \quad \forall i \in R^4, j \in N, \\
&\quad \forall v \in V, \forall t \in T, \forall \omega \in \Omega
\end{aligned} \tag{20}$$

$$\begin{aligned}
&\sum_{p \in P} \sum_{j \in R^3} \sum_{v \in V} \sum_{\substack{t' \in T \\ t' = t + a_{ij,v} + l_v}} x_{p,i,j,v,t',\omega} \\
&\quad + \sum_{p \in P} \sum_{j \in N} \sum_{v \in V} \sum_{\substack{t' \in T \\ t' = t + a_{ij,v} + l_v}} (x_{p,i,j,v,t',\omega}^{vmi} + x_{p,i,j,v,t',\omega}^{mto} \\
&\quad + x_{p,i,j,v,t',\omega}^{mts}) \leq c_i^{lod} \quad \forall i \in U, \forall t \in T, \forall \omega \in \Omega
\end{aligned} \tag{21}$$

$$h_{p,i,t,\omega} \leq b_{p,i}^{max_i^{vmi}} \quad \forall p \in P, \forall i \in N, \forall t \in T, \forall \omega \in \Omega \tag{22}$$

$$h_{p,i,t,\omega} \geq b_{p,i}^{min_i^{vmi}} \quad \forall p \in P, \forall i \in N, \forall t \in T, \forall \omega \in \Omega \tag{23}$$

$$u_{i,\omega}^{mto} \geq e^{mto} \times y_i^{mto} \quad \forall i \in N, \forall \omega \in \Omega \tag{24}$$

$$u_{i,\omega}^{mto} = 1 - \frac{\sum_{p \in P} \sum_{t \in T} s_{p,i,t,\omega}^{mto}}{\sum_{p \in P} \sum_{t \in T} d_{p,i,t,\omega}} \quad \forall i \in N, \forall \omega \in \Omega \tag{25}$$

$$y_i^{vmi}, y_i^{mto}, y_i^{mts} \in \{0, 1\} \quad \forall i \in N, \forall \omega \in \Omega \tag{26}$$

$$r_{p,i,t,\omega} \in \{0, 1\} \quad \forall p \in P, \forall i \in N, \forall t \in T, \forall \omega \in \Omega \tag{27}$$

$$q_{p,i,t,\omega} \geq 0 \quad \forall p \in P, \forall i \in U, \forall t, t' \in T, \forall \omega \in \Omega \tag{28}$$

$$\begin{aligned}
x_{p,i,j,v,t',\omega} &\geq 0 \quad \forall p \in P, \forall i \in R^4, \\
&\quad \forall j \in R^3, \forall v \in V, \forall t \in T, \forall \omega \in \Omega
\end{aligned} \tag{29}$$

$$\begin{aligned}
&x_{p,i,j,v,t',\omega}^{vmi}, x_{p,i,j,v,t',\omega}^{mto}, x_{p,i,j,v,t',\omega}^{mts} \geq 0 \\
&\quad \forall p \in P, \forall i \in R^4, \\
&\quad \forall j \in N, \forall v \in V, \forall t \in T, \forall \omega \in \Omega
\end{aligned} \tag{30}$$

$$h_{p,i,t,\omega} \geq 0 \quad \forall p \in P, \forall i \in R^5, \forall t, t' \in T, \forall \omega \in \Omega \tag{31}$$

$$s_{p,i,t,\omega}^{mto} \geq 0 \quad \forall p \in P, \forall i \in N, \forall t \in T, \forall \omega \in \Omega \tag{32}$$

$$u_{i,\omega}^{mto} \geq 0 \quad \forall i \in N, \forall \omega \in \Omega \tag{33}$$

$$d_{p,i,t,\omega}^{price} \geq 0 \quad \forall p \in P, \forall i \in N, \forall t \in T, \forall \omega \in \Omega \tag{34}$$

The objective function (2), combined with Equation (3), aims to maximise the expected profit by considering all revenue and operational costs, including production, inventory, transportation, backorder, order management, and setup costs. Constraints (4)-(6) represent the inventory balance at mills, cross docks, and warehouses. In Constraint set (7), the initial inventory at mills and warehouses is constrained to be limited to the average demand over the planning horizon. Constraint set (8) ensures

that the inventory at mills and external warehouses is at least equal to the initial inventory at the end of the planning horizon, thereby preventing the initial inventory from influencing production planning decisions. Constraint set (9) guarantees that the inventory level at customers served by the VMI strategy meets all demand on time. Constraint set (10) accounts for the backordered levels of customers served by the MTO strategy. Constraint set (11) treats any unmet demand from customers served by the MTS strategy as lost sales. Constraint set (12) introduces different demand functions for each customer, while Constraint set (13) determines whether the demand from customers served by the MTS strategy for a specific product is satisfied. Constraint set (14) determines the appropriate production-inventory strategy (MTS, MTO, or VMI) for each customer. Constraint set (15) ensures that the backordered demand of customers served by the MTO strategy is fulfilled by the end of the planning horizon. Constraints (16) and (17) represent the production capacity constraints at mills. Constraints (18)-(21) encompass the storage, transportation, and container loading constraints. Constraint sets (22) and (23) impose upper and lower bounds on the inventory level at customers served by the VMI strategy. Constraint set (24), along with Constraint set (25), ensures a minimum service level for customers served by the MTO strategy. Finally, Constraint sets (26)-(34) include binary and non-negativity constraints for the decision variables across each scenario.

5. Solution approach

The computational time and effort required to solve NP-hard problems with increasing problem size grow exponentially (Adeinat et al. 2022). Even for problems of moderate size, finding the optimal solution can be challenging, especially for non-concave structures like the model proposed in this paper. In addition, the inclusion of a two-stage scenario-based stochastic programming model further complicates the problem-solving process, making it even more challenging. Therefore, it becomes necessary to employ heuristics to generate nearly optimal solutions within a reasonable computational time.

In this study, we propose a metaheuristic approach based on the simulated annealing (SA) algorithm to tackle the formulated model. The distinctive qualities of the SA algorithm were the determining factors for its selection in solving the proposed model presented in this article. The SA method strikes a balance between diversification and intensification. It accomplishes this by combining two strategies: a random walk and iterative improvement. SA achieves intensification through its acceptance criterion. The probabilistic nature of

accepting worse solutions allows the algorithm to explore local regions effectively, preventing premature convergence to suboptimal solutions. SA also doesn't have to perform extensive evaluations of objective functions, unlike methods such as tabu search or any approach that assesses a neighbourhood. Besides, SA introduces diversification through the random walk strategy. By allowing the algorithm to occasionally accept worse solutions with a certain probability, SA can explore the solution space more extensively, preventing it from getting trapped in local optima. The SA algorithm has been successfully applied to various difficult combinatorial optimisation problems, as demonstrated by Yu and Lin (2014). By leveraging the benefits of SA, we aim to find high-quality solutions for the developed model efficiently.

In this study, we propose an SA-based metaheuristic approach to solve the two-stage stochastic MINLP model M_0 and determine the first-stage decisions. These decisions involve assigning customers to the MTS, MTO, and VMI strategies, as well as setting the sales prices for the products based on the assigned strategies.

The overall solution approach, as depicted in Figure 3, consists of generating a set of demand scenarios and then applying the SA algorithm for a fixed number of iterations to find the best solutions for the first-stage decision variables of model M_0 .

In addition, it is necessary to establish the deterministic equivalent versions of model M_0 as outlined below:

- M_1 The deterministic mixed-integer linear programming (MILP) equivalent of model M_0 assumes that the demand is deterministic, sales prices are parameters, and the assigned strategy to customers' demand is a variable.
- M_2 The deterministic mixed-integer linear programming (MILP) equivalent of model M_0 assumes that the demand is deterministic, while sales prices and the assigned strategy to customer's demand are parameters.

In our proposed approach, an initial solution is provided by solving model M_1 . At each iteration of the SA algorithm, the solutions for the first-stage variables are evaluated by solving the model M_2 for each demand scenario, separately. It is important to note that by treating the variables related to customers served under MTS, MTO, and VMI strategies, as well as sales prices (i.e. $y_i^{mts}, y_i^{mto}, y_i^{vmi}$, and $dep_{p,i}$), as parameters, the original two-stage stochastic MINLP model M_0 is transformed into an MILP model (M_2).

In the subsequent subsections, we provide a detailed discussion of the proposed solution approach, elaborating on its key components and mechanisms.

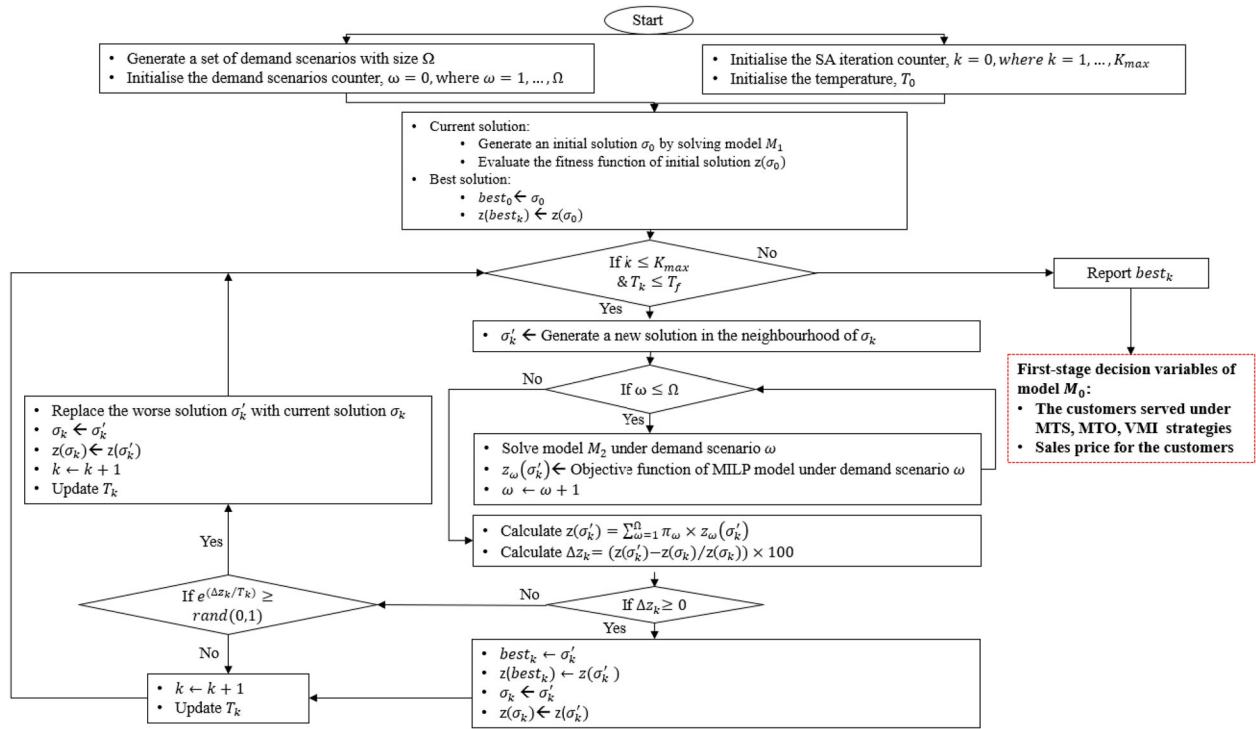


Figure 3. The SA-based metaheuristic solution approach to solve the proposed two-stage stochastic two-stage stochastic MINLP model.

5.1. Simulated annealing algorithm

In this study, we employ the SA algorithm to solve a maximisation problem. The SA algorithm, as a probabilistic search heuristic for maximisation problems, follows the basic structure outlined below (Adeinat et al. 2022):

- (1) The search heuristic begins by initialising the temperature and a feasible solution, denoted as x , with an objective value of $z(x)$.
- (2) A feasible neighbour solution, denoted as y , is probabilistically generated with an objective value of $z(y)$.
- (3) If $\Delta z = z(y) - z(x)$ is positive, the current best solution x is replaced with y . Otherwise, to prevent getting stuck in a local maximum, y is accepted based on a positive probability. The acceptance probability (P_k) is typically defined as $\exp(\Delta z / T_k)$, where T_k is a control parameter representing the temperature at the k^{th} iteration of the algorithm. The acceptance function indicates that a smaller value of ∇z is more likely to be accepted than a larger one. It also demonstrates that at higher temperatures, the probability of accepting worse solutions is higher, while the probability gradually decreases with decreasing temperature. If the value of the acceptance function is greater than a randomly generated number between 0 and 1, a worse solution is accepted.
- (4) The generation of new neighbours and the update of the current best solution continue until a

predetermined termination criterion is met, such as reaching a specified number of iterations or reaching a specified run time.

- (5) Upon meeting the termination criteria, the algorithm concludes and provides the current best solution, which represents a local optimal solution.

Table 1 shows the used notation in SA algorithm.

Table 1. Notation used in the proposed SA algorithm.

K_{max}	Maximum number of iterations
k	Current iteration number index, where $k = 1, \dots, K_{max}$
T_k	Temperature at iteration k , where $k = 1, \dots, K_{max}$
T_0	Initial temperature
T_f	Final temperature
λ	Cooling factor
P_k	Acceptance probability of a worse solution in iteration k , where $k = 1, \dots, K_{max}$
σ_0	Initial solution
σ_k	Current solution at iteration k , where $k = 1, \dots, K_{max}$
σ'_k	New solution generated at iteration k which is selected from the neighbouring solutions of σ_k , i.e. $\sigma'_k \in N(\sigma_k)$
z	Fitness function
z_ω	Fitness function under demand scenario ω , where $\omega = 1, \dots, \Omega$
$y_{i,k,j}$	1 if customer $i \in N$ is served under strategy j at iteration k , where $k = 1, \dots, K_{max}$ and $j = mts, mto, vmi$, 0 otherwise
$e_{p,i}^k$	Sales price of product $p \in P$ for customer $i \in N$ at iteration k , where $k = 1, \dots, K_{max}$
$e_{p,j}^{min}$	Minimum of sales price of product p under strategy j , where $j = mts, mto, vmi$
$e_{p,j}^{max}$	Maximum of sales price of product p under strategy j , where $j = mts, mto, vmi$
$best_k$	Best solution obtained so far up to iteration k , where $k = 1, \dots, K_{max}$

5.2. Algorithmic parameters in the proposed simulated annealing algorithm

When applying the SA algorithm to address a combinatorial optimisation problem, there are two types of decision to be made regarding algorithmic parameters: generic decisions that are applicable to all problems, and specific decisions tailored to the problem at hand (Radhakrishnan and Ventura 2000; Adeinat et al. 2022).

Generic decisions are applicable to any SA algorithm and are independent of the specific problem being solved. These decisions include determining the initial and final values for the temperature, selecting an acceptance function, choosing a cooling temperature function, and specifying the number of iterations. On the other hand, problem-specific decisions pertain to the characteristics of the problem itself and are highly dependent on its nature. These decisions involve choosing a suitable solution representation, determining methods for generating an initial feasible solution, and defining strategies for generating good neighbours from the current solution. Making informed choices for problem-specific decisions can lead to improved solution quality and reduced computational time. In the upcoming subsections, we will first discuss the generic choices before delving into the details of problem-specific decisions.

5.2.1. Generic decisions

In this section, we discuss the generic choices that need to be made when implementing the SA algorithm. These choices include selecting an acceptance function, determining the initial and final temperature values, deciding on a temperature cooling scheme, and specifying the number of iterations.

Acceptance function. This function determines the probability of accepting a worse solution during each iteration. The acceptance probability is calculated using Equation (35):

$$P_k = e^{\left(\frac{\Delta z_k}{T_k}\right)} = e^{\left(\frac{((z_k - z_{k-1})/z_{k-1}) \times 100}{T_k}\right)}, \quad (35)$$

where z_k represents the objective function value of the new solution, z_{k-1} represents the objective function value of the current solution, and Δz_k quantifies the percentage change between the new solution and the current solution. The temperature at iteration k is denoted by T_k .

Setting initial and final temperature. Various approaches have been proposed in the literature to determine the initial and final temperatures for the SA algorithm (e.g. Yu and Lin 2014; Li et al. 2013; Subramanian et al. 2013). However, in this paper, we adopt the methodology introduced by Adeinat et al. (2022). The initial temperature

is chosen such that there is a 90% probability of accepting a solution that is 10% worse than the current solution. Specifically, the initial temperature used in this study is 95, as indicated by Equation (36).

$$P_k = e^{(-10/T_0)} = 0.9 \rightarrow T_0 \approx 95 \quad (36)$$

Following the same strategy, the final temperature is set to 5, as described by Equation (37):

$$P_k = e^{(-10/T_f)} = 0.1 \rightarrow T_f \approx 5 \quad (37)$$

Temperature cooling scheme. To decrease the temperature at each iteration, we employ the logarithmic cooling scheme, which has been utilised in various supply chain problems (e.g. Subramanian et al. 2013; Franzin and Stützle 2019; Adeinat et al. 2022). The cooling factor λ is determined by Equation (38), and the temperature at iteration k is calculated using Equation (39):

$$\lambda = \frac{T_0 - T_f}{K_{max} \times T_0 \times T_f} \quad (38)$$

$$T_k = \frac{T_{k-1}}{1 + \lambda \times T_{k-1}} \quad \forall k, k = 1, \dots, K_{max} \quad (39)$$

Number of iterations. After conducting experiments on computational run time, the SA algorithm proposed in this paper is terminated after 100 iterations, denoted as $K_{max} = 100$. The solution obtained at the final iteration is considered the best-known solution for the given problem.

5.2.2. Problem-specific choices

In this subsection, we delve into the problem-specific choices, which include representing the solution, evaluating the fitness function, and generating initial and neighbouring solutions. These aspects are crucial in the context of the problem at hand.

Representing the solution. In this study, we present a solution representation structure depicted in Figure 4. Our approach involves constructing a matrix of size $(3 + m) \times n$, where n represents the number of customers and m represents the number of products. The first three rows of the solution structure correspond to the assignment of customers to the VMI, MTO, and MTS strategies. The remaining rows indicate the sales price of each product for each customer.

Evaluating fitness function. In accordance with the proposed solution approach, variables associated with customers served under MTS, MTO, and VMI strategies, as well as sales prices (i.e. y_i^{mts} , y_i^{mto} , y_i^{vmi} , and $and_{p,i}$) in the developed two-stage stochastic MINLP model M_0 are

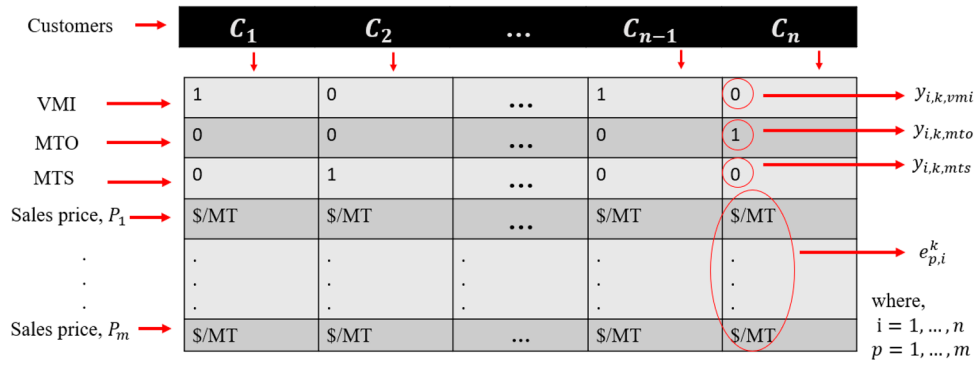


Figure 4. The solution representation in the proposed SA algorithm at iteration k .

transformed into parameters. As a result of this transformation, constraint set (14) becomes unnecessary. Consequently, in each iteration of the proposed SA algorithm, we solve a deterministic MILP model (M_2) for each demand scenario, separately. It is important to note that M_2 is solved for each scenario with a distinct demand pattern. To address potential infeasibility in M_2 due to unmet demand for VMI and MTO customers, we introduce two new variables as follows:

- $ud_{p,i,t}^{vmi}$ Quantity of unmet demand of product $p \in P$ at customer $i \in N$ served under VMI strategy at period $t \in T$
- $ud_{p,i,t}^{mto}$ Quantity of unmet demand of product $p \in P$ at customer $i \in N$ served under MTO strategy at period $t \in T$

Consequently, the updated versions of Constraint Sets (9), (10), and (23) are as follows, respectively:

$$h_{p,i,t} = h_{p,i,t-1} + \sum_{j \in R^4} \sum_{v \in V} \sum_{t' \in T} x_{p,j,i,v,t'}^{vmi} - d_{p,i,t} \times y_i^{vmi} + ud_{p,i,t}^{vmi} \quad \forall i \in N, \forall p \in P, \forall t \in T \quad (40)$$

$$\sum_{j \in R^4} \sum_{v \in V} \sum_{t' \in T} x_{p,j,i,v,t'}^{mto} - s_{p,i,t-1}^{mto} = d_{p,i,t} \times y_i^{mto} - s_{p,i,t}^{mto} - ud_{p,i,t}^{mto} \quad \forall i \in N, \forall p \in P, \forall t \in T \quad (41)$$

$$h_{p,i,t} + ud_{p,i,t}^{vmi} \geq b_{p,i}^{min_{vmi}} \quad \forall p \in P, \forall i \in N, \forall t \in T \quad (42)$$

It should be noted that model M_2 is deterministic, which means that the scenario indices have been removed from the group of variables.

To manage the additional costs resulting from unmet demand of VMI and MTO customers, we introduce two

new parameters as follows (large values are assigned to both parameters):

- $g_{p,i,t}^{vmi}$ Cost of unmet demand of product $p \in P$ at customer $i \in N$ served under VMI strategy at period $t \in T$
- $g_{p,i,t}^{mto}$ Cost of unmet demand of product $p \in P$ at customer $i \in N$ served under MTO strategy at period $t \in T$

In this regard, the objective function of model M_2 is defined by replacing Equations (2) and (3) with Equation (43):

$$\begin{aligned} & \text{Maximise } z_{\omega}(\sigma'_k) \\ &= \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in N} \sum_{v \in V} \sum_{t, t' \in T} e_{p,j} \times x_{p,i,j,v,t'}^{vmi} \\ &+ \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in N} \sum_{v \in V} \sum_{t, t' \in T} e_{p,j} \times x_{p,i,j,v,t'}^{mto} \\ &+ \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in N} \sum_{v \in V} \sum_{t' \in T} \sum_{t \in T^{mts}} e_{p,j} \times x_{p,i,j,v,t',t}^{mts} \\ &- \sum_{p \in P} \sum_{i \in U} \sum_{t \in T} g_{p,i}^{prd} \times q_{p,i,t} - \sum_{i \in R^5} \sum_{p \in P} \sum_{t \in T} g_i^{inv} \\ &\times h_{p,i,t} - \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in R^3} \sum_{v \in V} \sum_{t, t' \in T} g_{i,j,v}^{trs} \times x_{p,i,j,v,t',t} \\ &- \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in N} \sum_{v \in V} \sum_{t, t' \in T} g_{i,j,v}^{trs} \times x_{p,i,j,v,t',t}^{vmi} \\ &- \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in N} \sum_{v \in V} \sum_{t, t' \in T} g_{i,j,v}^{trs} \times x_{p,i,j,v,t',t}^{mto} \\ &- \sum_{p \in P} \sum_{i \in R^4} \sum_{j \in N} \sum_{v \in V} \sum_{t' \in T} \sum_{t \in T^{mts}} g_{i,j,v}^{trs} \times x_{p,i,j,v,t',t}^{mts} \\ &- \sum_{p \in P} \sum_{i \in N} \sum_{t \in T} w_p^{mto} \times s_{p,i,t}^{mto} - \sum_{p \in P} \sum_{l \in R^4} \sum_{j \in N} \sum_{t, t' \in T} o \end{aligned}$$

$$\begin{aligned}
& \times x_{p,l,j,v,t',t}^{mto} - \sum_{p \in P} \sum_{i \in U} \sum_{t \in T} s_{p,i} \times q_{p,i,t} \\
& - \sum_{p \in P} \sum_{i \in N} \sum_{t \in T} g_{p,i,t}^{vmi} \times u d_{p,i,t}^{vmi} - \sum_{p \in P} \sum_{i \in N} \sum_{t \in T} g_{p,i,t}^{mto} \\
& \times u d_{p,i,t}^{mto} \quad (43)
\end{aligned}$$

The fitness function of a newly generated solution at iteration k is calculated using Equation (44):

$$z(\sigma'_k) = \sum_{\omega=1}^{\Omega} \pi_{\omega} \times z_{\omega}(\sigma'_k) \quad \forall k, k = 1, I, K_{max} \quad (44)$$

Generating initial solution. In the SA algorithm, an initial feasible solution needs to be generated, as well as neighbouring solutions for the current solution at each iteration of the algorithm. The binary variables related to the production-inventory strategy assigned to the customers can take values of 0 or 1, while the sales price variables can take values ranging from 0 to infinity. The initial solution can be generated randomly, but in this study, we consider lower and upper bounds for the sales price values based on sales experts' opinions or the prices that customers are willing to pay in the current market (see, Phillips 2021). The sales price for customers is directly linked to the assigned strategy. In practice, customers served under the VMI strategy have higher sales prices compared to customers served under the MTO strategy for the same product, as they receive better services. Similarly, customers served under the MTS strategy pay less than MTO customers. When generating the initial solution, we do not have prior knowledge about the strategy that customers will be served. Therefore, the sales price of product p for customer i at iteration k , $e_{p,i}^k$, is randomly generated between $e_{p,j}^{min}$ and $e_{p,j}^{max}$ where $j = mts, mto, vmi$. To generate an initial solution for the binary variables related to customers served under each strategy, we solve model M_1 , which is the deterministic equivalent of model M_0 . This is done by considering the randomly generated sales prices as parameters and assuming a deterministic demand parameter equal to the average of all demand scenarios.

Generating neighbouring solutions. Based on the proposed solution structure for the developed SA algorithm, the swap operation is employed to generate neighbouring solutions for the assigned production-inventory strategy to the customers, specifically for the binary variables. Each customer can be served under only one strategy, such as MTS, MTO, or VMI. To accomplish this, the algorithm randomly selects a customer, and the values associated with two strategies are swapped randomly, effectively changing the assigned strategy for that customer.

Subsequently, the algorithm needs to generate values for the sales price variables, which are continuous variables. At each iteration k , the sales price values are adjusted to increase diversity when the temperature is high, and to focus on intensification when the temperature is low. To generate the sales prices at each iteration of the SA algorithm, we follow the methodology proposed by Adeinat et al. (2022). At iteration k , the step size $\tau_{p,i}^k$ for changing the sales price of product p for customer i , who is served under strategy j , is determined by Equations (45) and (46):

$$\nabla_{i,p,k} = (e_{p,j}^{max} - e_{p,j}^{min}) \times \frac{1}{n} \times T_k \quad (45)$$

$$\tau_{p,i}^k = - \left(\frac{\nabla_{i,p,k}}{2} \right) + (\nabla_{i,p,k} \times rand(0, 1)) \quad (46)$$

The step size, denoted as $\tau_{p,i}^k$, is defined as a function of the temperature at iteration k , T_k . It has larger values for higher temperatures and smaller values for lower temperatures. This approach ensures that the step size adapts to the current temperature, allowing for more significant changes in the sales price when the temperature is high, and smaller adjustments when the temperature is low. Consequently, the sales price for product p of customer i at iteration k is calculated using Equation (47):

$$e_{p,i}^k = \max(\min((\tau_{p,i}^k + e_{p,i}^{k-1}), e_{p,j}^{max}), \sum_j y_{i,k,j} \times e_{p,j}^{min}) \quad (47)$$

The process of generating a neighbouring solution is illustrated in Figure 5. In the first step, Customer C2 is chosen. Moving on to Step 2, the swap operation is executed, with Customer C2 transitioning from MTS to VMI strategy. In Step 3, the step size ($\tau_{p,i}^k$) is computed, and subsequently, in the final step, the sales price for each product for each customer ($e_{p,i}^k$) is determined.

6. Experimental results and discussion

In this section, we demonstrate the practical implementation of the proposed model and the SA algorithm through a real case study. The metaheuristic SA-based algorithm was coded in Python 3.7 and CPLEX solver version 12.10 was utilised to solve models M_1 and M_2 . The computational calculations were performed on a Windows 11 operating system, running on an Intel® Xeon® 2.26 GHz with 100GB RAM. Below, we outline the details of our case study, and subsequently, we establish the experimental design for our study.

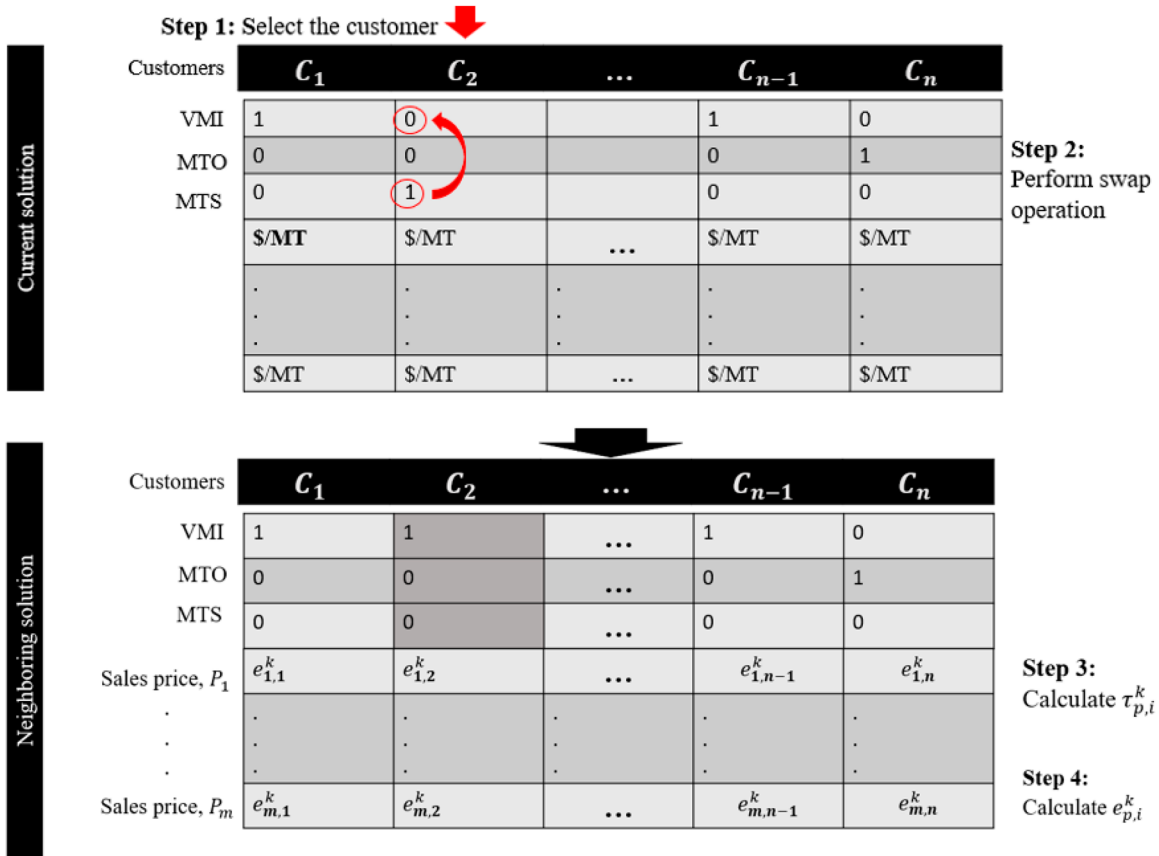


Figure 5. The procedure for generating a neighbouring solution.

6.1. Case study

The case study conducted in this research focuses on a company operating in the pulp and paper industry in Canada. The company has a diverse product portfolio, including publication paper, containerboard, and packaging products. Their customer base spans across various regions globally. For this study, we specifically concentrate on 37 customers located in North America, encompassing both Canada and the United States.

The company operates three primary production units (mills), all situated in Canada. Each mill has its designated storage facilities and production capacities. In addition, the company has the capability to fulfil customer demand from 20 external warehouses strategically located across Canada and the United States. Delivery to customers is carried out using either rail or truck transportation. The supply network is further supported by four cross docks, facilitating the consolidation of products from different mills for efficient delivery to customers or external warehouses.

While the company encounters seasonal demand variations, the operational planning, including production, inventory control, and distribution, is performed on a weekly basis. However, crucial decisions regarding sales prices and the assignment of production-inventory

strategies to customers are based on the forecasted demand available for the entire year.

Figure 6 presents an illustration of the supply chain configuration considered in this study, serving as the specific case study scenario.

6.2. Demand scenarios generation

To accurately capture the real-world dynamics of customer demand in this study, it is essential to incorporate different demand patterns for each season. However, accounting for a large number of scenarios can significantly increase the computational complexity of solving the problem. Conversely, an insufficient number of scenarios may result in unreliable recommendations regarding the assignment of production-inventory strategies to customers. To strike a balance, we have conducted experiments and devised the following steps to generate the demand scenarios:

Step 1: Data Collection. The actual data is collected for each customer's seasonally aggregated demand over a one-year period, encompassing four seasons.

Step 2: Probability distribution determination. Based on the available actual data, the top four probability distributions are identified using the residual sum of squares (RSS) as the evaluation metric. These

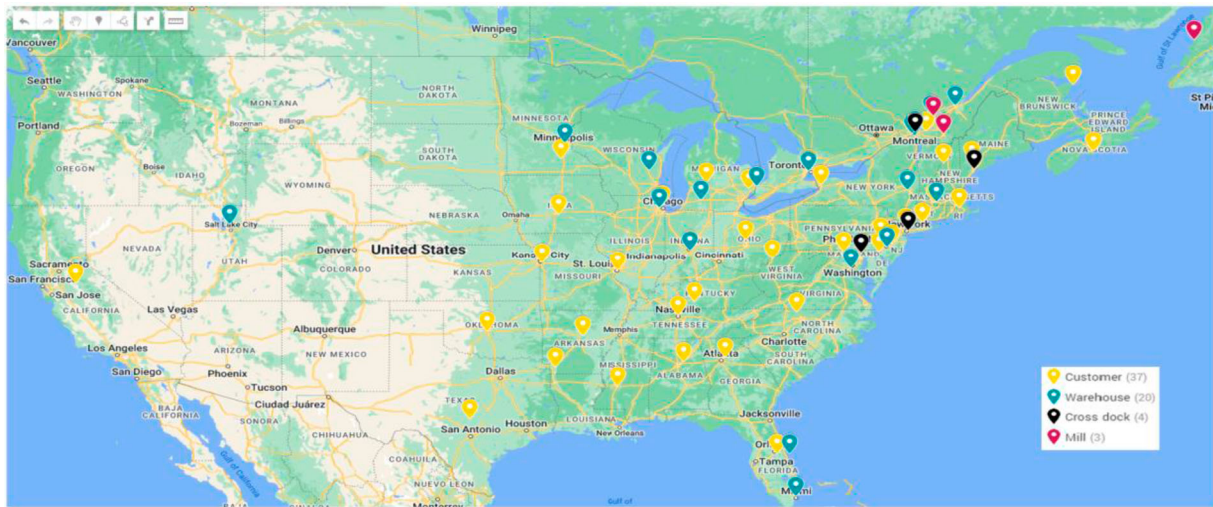


Figure 6. Configurations of the considered case study from a real pulp and paper supply chain (map source: Google Maps).

distributions represent the best fit to the observed demand data.

Step 3: Demand pattern generation. Once the four best probability distributions are determined for each customer, four demand patterns are generated for each season, resulting in a total of 16 demand patterns. These patterns capture the variations and characteristics observed in the historical data.

Step 4: Scenario Creation. By aggregating the generated seasonal demand patterns for all customers, a set of 16 demand scenarios is created. These scenarios are labelled as S1-S4 for season 1, S5-S8 for season 2, S9-S12 for season 3, and S13-S16 for season 4. Each scenario represents a distinct combination of demand patterns across customers, providing a representation of the possible demand fluctuations.

By following these steps, we are able to generate a set of demand scenarios that capture the potential fluctuations in demand. These scenarios provide a representation of the various demand patterns observed in the actual data. This allows us to evaluate and analyse the effectiveness and resilience of the proposed production-inventory strategies under diverse and realistic demand conditions.

Table 2 presents the average RSS values for the generated demand patterns in each season. These RSS values serve as an indicator of how well the generated patterns fit the actual data. Thus, demand pattern #1 exhibits the closest fit to the actual data, indicating a higher level of accuracy in representing the demand fluctuations. On the other hand, demand pattern #4 demonstrates the least fit, suggesting a lower level of accuracy in capturing the actual demand pattern.

In addition, Figure 7 provides a visual comparison between the actual data and the generated demand patterns (#1-#4) for each season. This schematic

Table 2. The average residual sum of squares (RSS) for the generated demand patterns in each season.

Season	Demand pattern #			
	#1	#2	#3	#4
Season 1	1.606	1.734	2.036	2.084
Season 2	1.566	1.705	2.045	2.090
Season 3	1.380	1.662	1.826	1.888
Season 4	1.830	1.975	2.333	2.368

Table 3. The quality of the generated data in each scenario.

Season	Scenario	Market size (MT)*	Market size change (%)	Demand inaccuracy (%)
Season 1	Actual	198,653	—	—
	S1	193,843	−2.42	[−%10, +%10]
	S2	199,264	+0.31	[−%20, +%20]
	S3	199,413	+0.38	[−%30, +%30]
Season 2	S4	202,054	+1.71	[−%40, +%40]
	Actual	206,024	—	—
	S5	203,399	−1.27	[−%10, +%10]
	S6	203,897	−1.03	[−%20, +%20]
Season 3	S7	208,064	+0.99	[−%30, +%30]
	S8	211,007	+2.42	[−%40, +%40]
	Actual	207,411	—	—
	S9	207,321	−0.04	[−%10, +%10]
Season 4	S10	203,764	−1.76	[−%20, +%20]
	S11	208,008	+0.29	[−%30, +%30]
	S12	212,570	+2.49	[−%40, +%40]
	Actual	211,259	—	—
Season 4	S13	208,571	−1.27	[−%10, +%10]
	S14	210,424	−0.4	[−%20, +%20]
	S15	207,331	−1.86	[−%30, +%30]
	S16	213,998	+1.30	[−%40, +%40]

*MT: metric tons

representation allows for a qualitative assessment of the similarity and alignment between the generated patterns and the actual demand data. Table 3 presents additional information that assesses the quality and accuracy of the generated data in each scenario compared to the actual data. To capture a diverse range of inaccuracies, we made some modifications to the data in each scenario.

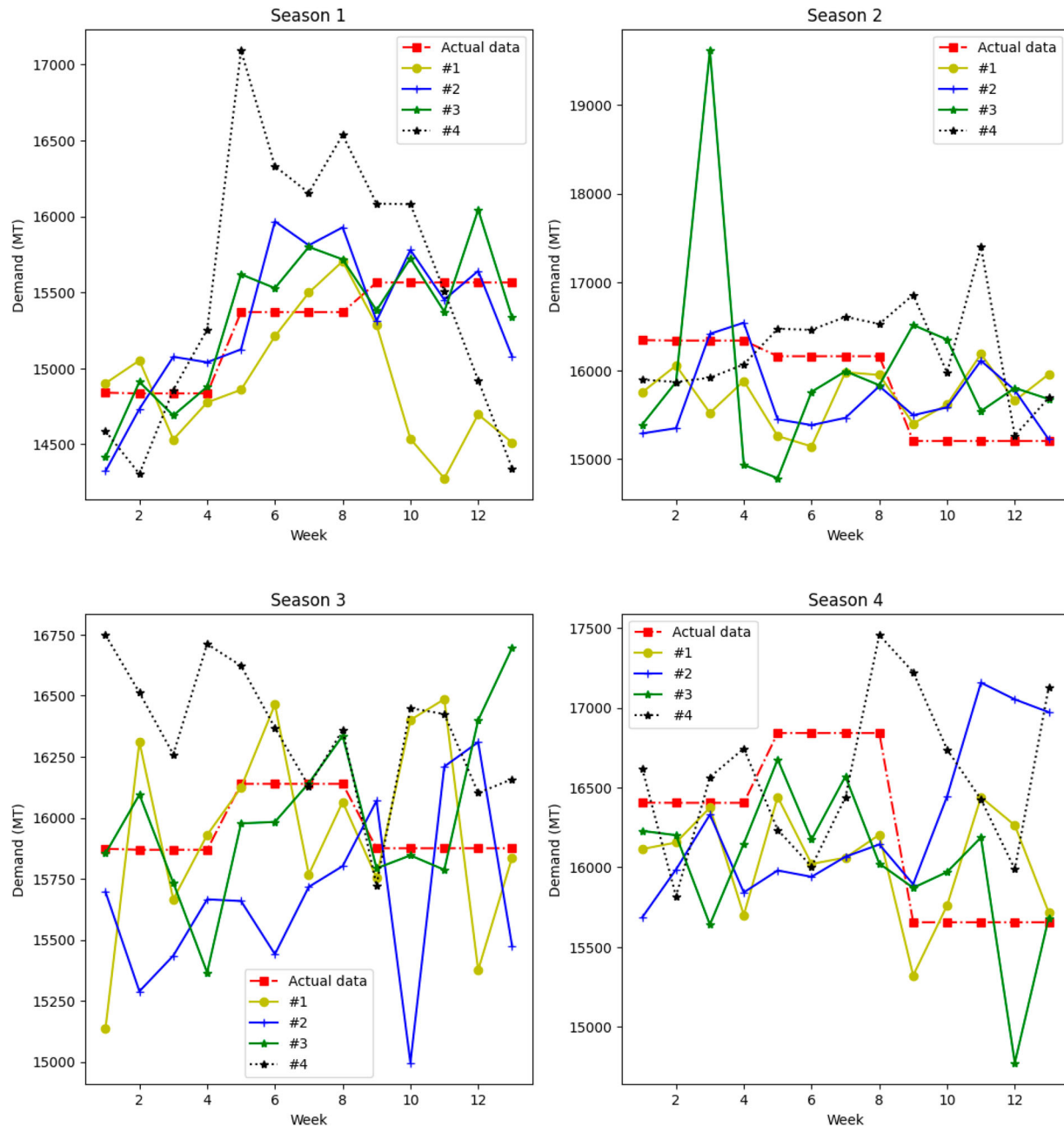


Figure 7. Generated five demand patterns based on the actual seasonal data (MT: metric tons).

Furthermore, Table 3 illustrates the disparity in market size between the generated demand patterns and the corresponding actual seasonal demand values. While our primary objective is to keep a similar total market size across different scenarios, it is important to acknowledge that within each scenario, certain customers may experience overestimation or underestimation.

6.3. Computational results and discussion

Solving problem instances based on real-world requirements is a crucial aspect in the literature of supply

chain planning (Brandenburg et al. 2014). The developed two-stage stochastic MINLP model M_0 for the case study considered in this study under one scenario, consists of 102,672 constraints, 2,331 binary variables and 6,184,427 continuous variables. As the number of scenarios increases, the number of decision variables (excluding the first-stage variables) and constraints grows linearly. The computation time, however, is expected to increase exponentially (Dyer and Stougie 2006; Weskamp et al. 2019). It is worth noting that even linear relaxation of model M_0 cannot be solved using CPLEX with 16 scenarios due to out-of-memory error. Given this situation,

it is evident that solving the nonlinear model M_0 becomes impossible.

On the other hand, even with the consideration of 16 scenarios in this study, solving the model M_0 using the proposed SA-based metaheuristic solution approach, limited to 100 iterations, still requires approximately 15 days. Given the considerable time required for solving the problem, it is impractical to consider a larger number of scenarios. Therefore, we have restricted our study to only 16 scenarios and limited the SA iteration to 100.

Table 4 displays the best expected profit achieved, the number of customers assigned to each production-inventory strategy, and the average sales price under each strategy. These results were obtained through 100 iterations of the SA algorithm. It is important to note that we have taken into account the same demand parameters (i.e. a and b) for all customers served under various production-inventory strategies.

Results indicate that the number of considered scenarios has an impact on the first-stage decisions, which include the assigned production-inventory strategy and sales price strategy, as well as the overall profit. Results also demonstrate that the implementation of the SA algorithm can enhance the initial profit across different numbers of scenario. It is important to emphasize that the initial profit for each number of scenarios was derived by solving the model M_2 , incorporating an initial solution for the first-stage decisions for each individual scenario, and subsequently calculating the average. By implementing the SA algorithm, the expected profit was improved by up to 1.43%, equivalent to \$440,851 under 12 scenarios.

To assess the performance of the best solution obtained under different numbers of scenarios, we examined these solutions individually in each scenario. Results are reported in Table 5. Based on the observations, the solution found under 16 scenarios does not yield

Table 4. Quality of the best solution found by the SA algorithm under varying numbers of scenario in terms of profit and the first-stage decisions (initial strategy: 16|15|6¹, initial average of sales price: 724|624|543²).

# Scenario	1	4	8	12	16
Scenarios included	–	S1 S5 S9 S13	S1, S2 S5, S6 S9, S10 S13, S14	S1-S3 S5-S7 S9-S11 S13-S15	S1-S4 S5-S8 S9-S12 S13-S16
Computation time	21 h	3.7 days	7.4 days	11.4 days	14.9 days
Initial Profit (\$)	31,386,480	30,705,257	30,768,867	30,806,634	31,042,106
Best Profit (\$)	31,622,681	31,021,884	31,176,801	31,247,485	31,395,876
Improvement (\$)	236,200.74	316,626	407,934	440,851	353,770
Improvement (%)	0.75	1.03	1.33	1.43	1.14
Best strategy	14 15 8	15 14 8	15 15 7	15 14 8	16 12 9
Best avg. sales price	721 624 554	723 626 554	723 629 554	721 622 554	719 625 554

¹(#VMI|#MTO|#MTS)

²(VMI(\$/MT)|MTO (\$/MT)|MTS(\$/MT))

Table 5. Evaluation of solutions found through different numbers of scenario.

Scenarios	# Scenario				
	1	4	8	12	16
S1	29,728,653*	29,836,197	29,911,900	29,876,569	29,769,090
S2	30,320,071	30,381,041	30,508,018	30,433,129	30,360,000
S3	29,504,726	29,683,304	29,707,053	29,723,638	29,625,735
S4	30,132,160	30,224,634	30,320,909	30,297,274	30,198,811
S5	30,213,051	30,300,446	30,396,064	31,473,380	31,354,057
S6	31,006,217	31,109,072	31,183,334	31,164,472	31,053,482
S7	31,916,505	32,032,973	32,111,850	32,079,791	31,972,217
S8	31,386,748	31,504,346	31,568,488	31,544,088	31,450,380
S9	31,680,385	31,743,958	31,855,920	31,777,503	31,705,760
S10	31,185,351	31,235,656	31,352,037	31,282,605	31,205,803
S11	31,755,803	31,850,730	31,942,324	31,860,110	31,775,300
S12	32,325,945	32,366,440	32,504,610	32,428,584	32,362,014
S13	32,147,945	32,206,936	32,327,261	32,208,964	32,183,017
S14	31,699,145	31,775,347	31,879,879	31,772,344	31,754,005
S15	31,227,605	31,284,890	31,397,739	31,317,327	31,268,318
S16	34,200,118	34,286,743	34,405,391	34,297,114	34,296,029
Avg. (S1-S16)	31,276,902	31,363,920	31,460,798	31,471,056	31,395,876
Improvement (\$)	–	87,018	96,879	10,257.23	(75,180)
Improvement (%)	–	0.28%	0.31%	0.03%	(0.24) %

* Total profits are reported in dollars (\$).

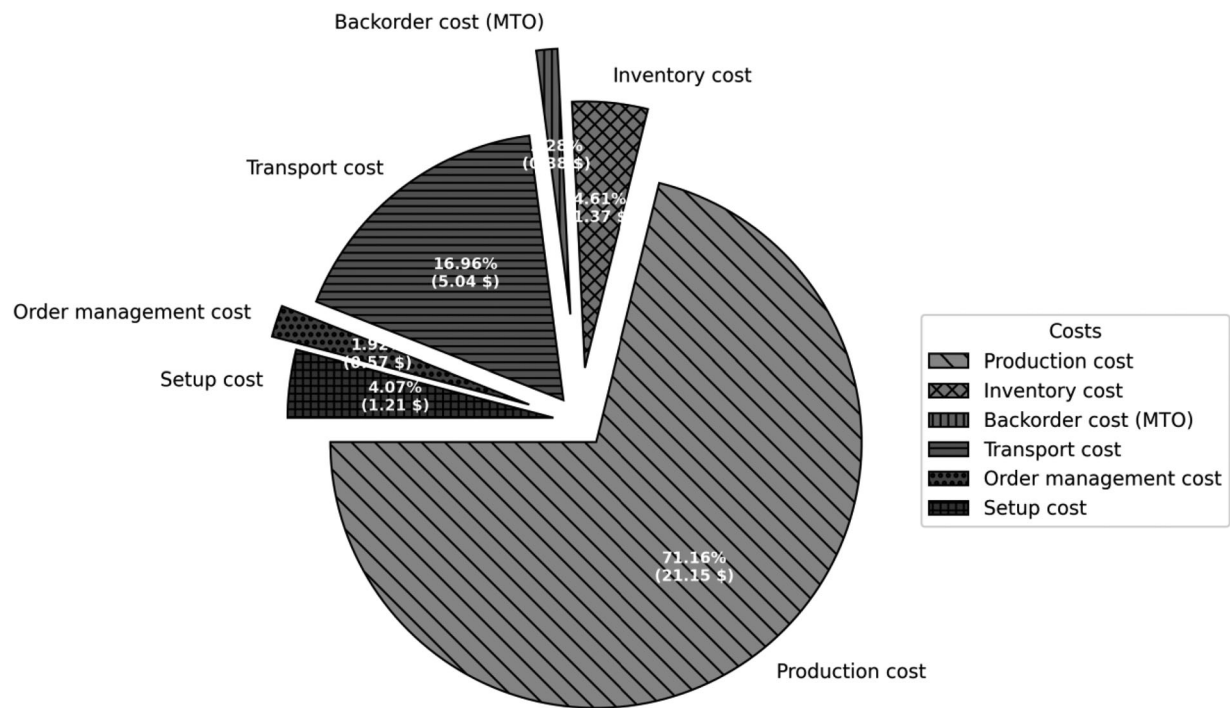


Figure 8. The distribution of average costs (million \$) across 12 scenarios.

better results when solving each scenario individually. This suggests that the proposed algorithm may require more iterations or computation time when the number of scenarios exceeds 12. Therefore, we have selected the solution found under 12 scenarios for further analysis.

Considering the initial solutions for the first-stage variables, the number of customers assigned to the VMI, MTO, and MTS strategies was 16, 15, and 6, respectively. However, after finding the best solution by considering 12 scenarios, these numbers changed to 15, 14, and 8, respectively. Despite the algorithm suggesting a decrease in the number of customers served under the MTO and VMI strategies when considering all 12 scenarios, it is possible to achieve a better profit (with a 1.43% improvement) by individually adjusting the sales prices for the customers. Notably, even the average sales prices found under the 12 scenarios show a decrease in the average sales price for VMI customers from 724 to 721, and a change for MTO customers from 624 to 622. The results highlight the significance of individually adjusting the sales price for each customer.

Figure 8 additionally visualises the distribution of average costs among the scenarios encompassed within the set of 12 scenarios. As depicted in the figure, the most significant expenses are associated with production and transportation costs.

Analysing the assigned strategies and determined sales prices for each customer under different numbers of

scenario, as depicted in Table 6, offers valuable insights. Results highlight that, although the overall number of customers served under each strategy remains relatively consistent, there are significant variations in the strategies assigned to individual customers. This variability can be attributed to the varying levels of inaccuracy for different customers across different scenarios. Furthermore, it is noteworthy that the market size exhibits minor fluctuations across the scenarios, as illustrated in Table 3. This stability in market size contributes to the similarity in the number of customers served under each strategy.

The highlighted rows in Table 6 indicate the customers for whom there have been changes in the assigned production-inventory strategies when considering different number of scenarios. As the number of scenarios increases, fewer customers experience changes in their assigned strategies. In other words, when considering 12 scenarios, the strategy assigned to 5 out of 37 customers (approximately 14%) has been changed, whereas with 4 scenarios, this value is 8 out of 37 (approximately 22%). Hence, there is a convergence in the assigned strategies for the customers as the number of scenarios increases. However, in terms of sales price, customers may experience fluctuations even when they are assigned the same strategy. This pattern remains consistent across different numbers of scenarios. Excluding the customers whose sales prices were changed due to strategy modifications, the remaining customers may encounter price increases

Table 6. The first-stage decisions including the production-inventory strategy and the average of sales price for the customers when varying the number of scenarios.

#Scenarios	Initial		1		4		8		12	
Customers	Strategy	Avg. price (\$/MT ¹)	Strategy	Avg. price (\$/MT)	Strategy	Avg. price (\$/MT)	Strategy	Avg. price (\$/MT)	Strategy	Avg. price (\$/MT)
C1	VMI	725	VMI	719 ▼	VMI	716 ▼	VMI	715 ▼	VMI	719 ▲
C2	VMI	723	VMI	724 ▲	VMI	726 ▲	VMI	730 ▲	VMI	719 ▼
C3	VMI	720	VMI	721 ▲	VMI	727 ▲	VMI	727 ➡	VMI	732 ▲
C4	VMI	722	VMI	723 ▲	VMI	713 ▼	VMI	726 ▲	VMI	725 ▼
C5	MTO	625	MTO	619 ▼	MTO	628 ▲	MTO	630 ▲	MTO	636 ▲
C6	VMI	724	VMI	727 ▲	VMI	724 ▼	VMI	725 ▲	VMI	729 ▲
C7	VMI	725	VMI	720 ▼	VMI	713 ▼	VMI	719 ▲	VMI	728 ▲
C8	MTO	627	MTO	626 ▼	MTO	620 ▼	MTO	637 ▲	MTO	615 ▼
C9	VMI	721	VMI	726 ▲	VMI	730 ▲	VMI	720 ▼	MTO ²	628
C10	VMI	714	MTO	607	MTS	554	VMI	718	VMI	718 ➡
C11	VMI	732	VMI	718 ▼	VMI	718 ➡	VMI	730 ▲	VMI	716 ▼
C12	VMI	731	VMI	719 ▼	VMI	720 ▲	VMI	719 ▼	VMI	723 ▲
C13	VMI	722	VMI	727 ▲	VMI	726 ▼	VMI	721 ▼	VMI	718 ▼
C14	VMI	722	VMI	719 ▼	VMI	721 ▲	VMI	724 ▲	VMI	713 ▼
C15	MTO	628	MTO	625 ▼	MTO	635 ▲	MTO	632 ▼	MTS	554
C16	MTO	621	MTO	630 ▲	MTO	616 ▼	MTO	620 ▲	MTO	629 ▲
C17	VMI	722	VMI	715 ▼	VMI	725 ▲	VMI	724 ▼	VMI	727 ▲
C18	VMI	719	VMI	719 ➡	VMI	732 ▲	VMI	737 ▲	VMI	715 ▼
C19	VMI	733	MTS	554	VMI	731	MTO	632	VMI	720
C20	VMI	729	VMI	725 ▼	VMI	728 ▲	VMI	713 ▼	VMI	717 ▲
C21	MTO	630	MTO	632 ▲	MTO	634 ▲	MTO	634 ➡	MTO	633 ▼
C22	MTO	623	MTO	628 ▲	MTO	636 ▲	MTO	636 ➡	MTO	639 ▲
C23	MTO	623	MTO	619 ▼	MTO	626 ▲	MTO	625 ▼	MTO	625 ➡
C24	MTO	637	MTO	618 ▼	MTS	554	MTO	624	MTO	632 ▲
C25	MTO	633	MTO	630 ▼	MTO	618 ▼	MTO	630 ▲	MTO	628 ▼
C26	MTS	552	MTS	554 ▲	MTS	554 ➡	MTS	554 ➡	MTS	554 ➡
C27	MTS	537	MTS	554 ▲	MTS	554 ➡	MTS	554 ➡	MTS	554 ➡
C28	MTS	543	MTS	554 ▲	MTO	619	MTS	554	MTS	554 ➡
C29	MTO	622	MTO	635 ▲	MTS	554	MTO	640	MTO	631 ▼
C30	MTO	627	MTS	554	MTO	619	MTO	629 ▲	MTO	621 ▼
C31	MTS	548	MTS	554 ▲	MTS	554 ➡	MTS	554 ➡	MTS	554 ➡
C32	MTO	628	MTO	626 ▼	MTO	617 ▼	MTO	621 ▲	MTS	554
C33	MTO	622	MTO	628 ▲	MTO	623 ▼	MTO	614 ▼	MTO	612 ▼
C34	MTS	551	MTS	554 ▲	MTS	554 ➡	MTS	554 ➡	MTS	554 ➡
C35	MTO	622	MTS	554	MTO	638	MTS	554	MTO	627
C36	MTS	543	MTO	609	MTS	554	MTS	554 ➡	MTS	554 ➡
C37	MTO	627	MTO	619 ▼	MTO	630 ➡	MTO	630 ➡	MTO	626 ▼

▼ Decrease; ▲ Increase; ➡ Unchanged

¹MT: metric tons²The highlighted rows indicate the alteration in the assigned strategy for customers across different scenarios considered.

of up to 1.5% (customer C16) and price decreases of up to 3.5% (customer C8) when considering 12 scenarios. By adjusting the sales price, specifically increasing it by 4.2% for five MTO customers and 1.1% for eight VMI customers and decreasing it by 7.6% for seven MTO customers and 8.5% for six VMI customers,

compared to the solution obtained under 8 scenarios, we achieved an improvement of 0.03% in the expected profit by solving the problem while considering 12 scenarios. Results also indicate that changing the assigned strategy for customers can lead to significant fluctuations in sales prices. For example, under 12 scenarios, customer C19

Table 7. Market size variation under different scenarios included in 12 scenarios (the highlighted rows indicate the alteration in the assigned strategy for customers across different scenarios considered).

	Inaccuracy in forecasting demand (%)												Avg. seasonal demand (metric tons)	Max. increase (%)	Min. decrease (%)	Assigned strategy under 12 scenarios
	S1	S2	S3	S5	S6	S7	S9	S10	S11	S13	S14	S15				
C1	-10	19	-24	-3	-9	28	9	21	-7	0	0	0	620	28	-24	VMI
C2	8	6	15	8	1	25	10	16	22	0	0	0	223	25	0	VMI
C3	10	-7	20	-4	-1	-10	-9	-18	29	8	13	18	2442	29	-18	VMI
C4	9	22	16	-8	18	26	-10	-2	2	5	-15	-18	1143	26	-18	VMI
C5	10	14	11	-2	-4	-4	8	-20	-1	-6	-19	19	9822	19	-20	MTO
C6	3	-8	20	1	-7	-9	-10	11	24	7	-7	6	1187	24	-10	VMI
C7	2	2	18	5	19	13	-4	0	-28	-8	-1	20	3159	20	-28	VMI
C8	9	22	-29	2	19	-29	-9	-9	-10	-3	17	-3	8322	22	-29	MTO
C9	0	0	0	0	0	0	0	0	0	-6	0	0	262	0	-6	MTO
C10	0	0	0	0	0	0	0	0	0	6	0	0	35	6	0	VMI
C11	10	19	27	-8	19	-26	-10	-17	11	-5	-5	18	1710	27	-26	VMI
C12	0	0	0	0	0	0	0	0	0	8	19	30	191	30	0	VMI
C13	10	16	24	-7	-3	31	-10	-19	16	-1	-19	29	1255	31	-19	VMI
C14	-1	-2	10	-2	1	-9	-10	-14	29	1	1	1	2208	29	-14	VMI
C15	10	-19	20	5	-18	3	9	0	0	-5	0	-3	15296	20	-19	MTS
C16	10	11	-21	-1	1	1	4	-4	-23	2	7	-15	10282	11	-23	MTO
C17	0	0	0	0	0	0	-4	0	0	10	-3	-19	641	10	-19	VMI
C18	10	11	-31	-10	-6	7	-6	9	28	-2	7	26	975	28	-31	VMI
C19	-6	-7	18	-10	-4	-10	-10	-15	-15	8	12	17	558	18	-15	VMI
C20	10	5	15	2	-13	-15	-10	-6	-6	-7	-5	1	1939	15	-15	VMI
C21	9	13	9	0	-6	12	-6	-4	-7	4	19	7	20498	19	-7	MTO
C22	8	8	23	-3	-8	5	2	3	7	-2	-1	8	19732	23	-8	MTO
C23	5	20	-5	3	13	-19	-3	20	23	-9	-17	3	941	23	-19	MTO
C24	10	17	-1	-3	7	1	8	14	17	-2	-1	-1	2248	17	-3	MTO
C25	9	10	27	10	-3	-7	3	3	-23	6	-3	15	885	27	-23	MTO
C26	10	11	7	-4	15	-2	3	12	-24	-2	-15	-12	15667	15	-24	MTS
C27	9	9	8	-1	-5	-2	1	-4	3	-9	-9	-7	20733	9	-9	MTS
C28	9	11	-1	1	-1	-4	-5	-5	-3	-7	-9	8	11613	11	-9	MTS
C29	-3	15	11	-2	-2	5	1	3	24	4	-3	-14	22052	24	-14	MTO
C30	3	5	5	8	12	14	0	-1	29	0	1	0	2347	29	-1	MTO
C31	-4	-10	12	0	8	5	-10	2	30	5	0	32	2548	32	-10	MTS
C32	0	0	0	0	0	0	3	-19	27	7	-12	30	691	30	-19	MTS
C33	-5	2	5	-6	16	3	2	-2	9	-9	1	-18	350	16	-18	MTO
C34	4	1	2	38	4	63	0	0	0	7	16	22	48	63	0	MTS
C35	-10	18	7	-1	-1	-5	-2	-3	-3	8	8	-12	15691	18	-12	MTO
C36	10	-3	25	-10	-20	27	-2	19	-1	0	-1	-3	935	27	-20	MTS
C37	-6	-6	4	-3	20	19	10	0	-13	-8	20	-6	6592	20	-13	MTO

experiences a 13.9% increase in sales price when transitioning from MTO to VMI strategy, while customer C9 sees a 12.8% decrease in sales price when switching from VMI to MTO strategy.

The analysis of the best solution found when considering 12 scenarios indicated that 15 VMI customers possessed 9% of the total demand, while 14 MTO customers accounted for 58% of the total demand, and eight (8) MTS customers held 33% of the total demand. This indicates that the VMI strategy was primarily allocated to customers with lower demand, while customers with higher demand were considered for the MTO strategy as long as the service level limitation permitted (it is assumed that 80% of MTO demands would be fulfilled within the specified timeframe).

Table 7 presents additional details concerning the inaccuracy level of forecasted demand for each customer

under different scenarios, aiming to provide a clearer understanding of the assigned production strategy to the customers. The customers who underwent at least one change in their production-inventory strategy under different numbers of scenarios are indicated by the highlighted rows in Table 7. Around 73% of the assigned strategies to 27 out of 37 customers have remained consistent despite considering various demand scenarios of varying accuracy. For the remaining 10 out of 37 customers, there is either a wide range of fluctuations in the forecasted demand or a substantial amount of demand exists for them. As an example, customer C15 possesses a demand of 15,691 metric tons (MT), which is comparatively high compared to the demands of other customers. In addition, there is a notable range of inaccuracies for that demand, spanning from -12 to +18. Hence, it is unsurprising that the strategy assigned to this

customer underwent changes when considering different scenarios.

7. Managerial insights

From the computational experiments conducted, several managerial insights can be highlighted. The case study focused on the company's aim to comprehend how to adapt production strategies for different customers and effectively coordinate sales price decisions. At present, the company does not possess a well-defined strategy to tackle this situation. Considering that the data used in this study closely approximates real data, these insights provide valuable points for the company to consider when making decisions regarding the allocation of production-inventory strategy and determining sales prices for customers at the tactical planning level, such as yearly planning:

- The conducted study provided the company with valuable insights concerning the applicability of MTS, MTO, and VMI strategies to potential customers. The results indicated that approximately 40% of customers, despite representing only 9% of the total market size, could be effectively served through the VMI strategy. On the other hand, MTO customers, constituting more than half of the market size, accounted for 38% of the customer base. In addition, the company can utilise its excess capacity to fulfil the requirements of 22% of customers, which corresponds to approximately 33% of the total demand. Moreover, the findings highlighted a preference for employing the VMI strategy for customers with lower demand, while the MTS strategy proved more suitable for customers with higher demand.
- In this study, we explored different levels of demand accuracy for customers and illustrated that the chosen production-inventory strategy for customers could be influenced, even with minimal fluctuations in the total market size. By considering additional scenarios, we could offer more robust decisions concerning the assigned strategy for customers. This underscores the significance of demand accuracy in determining production-inventory strategies. Specifically, we demonstrated that, as the magnitude of customer demand fluctuations increases, there is a corresponding increase in the variability of the assigned strategy for customers, particularly when the customer's demand is substantial.
- Another objective of this study was to highlight the coordination between sales price decisions and the strategy assigned to customer demand. As noted by Noori-Daryan, Taleizadeh, and Jolai (2019), pricing

plays a significant role in shaping market demand within supply chains. Consequently, it can alter the assigned production-inventory strategy and subsequently impact operational decisions related to production, inventory, and distribution activities. We demonstrated the impact that adjusting the sales price can have on the company's profitability.

- The model proposed in this study offers a valuable tool for facilitating the decision-making process within the company, particularly when considering changes to the assigned production-inventory strategy for customers due to various reasons. In practical settings, exploring different options can be challenging; however, with the utilisation of this model, testing and evaluating alternative approaches becomes considerably more accessible and straightforward.
- Moreover, the proposed optimisation model also offers the flexibility to alter the preassigned strategy for customers and to assess the resulting impact on the performance of the production system.

8. Conclusion

This research article was motivated by a real-world problem and focused on examining the coordination between sales price and assigned production-inventory strategies in a complex multi-level, multi-product supply chain. The considered supply chain comprises multiple customers who can be served under different strategies, including MTS, MTO, and VMI. Consequently, the sales price needs to be adjusted based on the assigned strategy to align with customer demand, as each strategy provides different service levels. The objective of the company under study is to determine which customers should be served under specific strategies and at what prices. According to Adeinat et al. (2022) the influence of price on product demand is evident, as pricing decisions directly impact product demand and subsequently affect upstream supply chain decisions, such as production, distribution, and inventory management. However, the coordination between sales price and production-inventory strategies is often investigated independently. In this article, a two-stage stochastic non-linear programming model was developed to address this coordination challenge. The model aimed to determine the customers to be served under each strategy and the corresponding sales price for each customer, considering different product demands. The study demonstrated that both the strategy assigned to customers and the sales price can be altered under various demand scenarios. The findings further exemplified how adjusting the sales price can lead to an increase in overall profit. The study presented evidence that, when the same strategy is considered, the

sales price can display minor variations across different demand scenarios. This fluctuation becomes particularly noteworthy when there is a change in the assigned production-inventory strategy for customers.

To tackle the computational complexity of the formulated problem, we introduced a metaheuristic algorithm employing Simulated Annealing (SA) to solve the presented model. Through the implementation of this solution approach, we were able to enhance the initial solution by considering 16 different scenarios. However, the results indicated that the algorithm achieved superior performance in finding solutions when considering 12 scenarios. This finding underscores the importance of evaluating the solution's quality across all scenarios. In cases with a larger number of scenarios, it implies that the proposed algorithm may require additional iterations beyond the initial 100 iterations to attain better solutions.

The model proposed in this study demonstrates promising applicability in real-world applications across a wide range of industries. It holds particular relevance for businesses that serve diverse customers and offer a variety of products where demand is highly sensitive to fluctuations in sales price. Future research in this field could explore extending the model to incorporate different price-sensitive demand functions for various strategies and customer segments. Investigating the impact of price response demand function parameters, similar to the study conducted by Adeinat et al. (2022), on sales price and assigned strategies would also be valuable. In addition, the quality of solutions provided by a two-stage stochastic programming model relies on the number of scenarios considered in the analysis. However, as highlighted by Weskamp et al. (2019), the complexity of the model formulation is significantly influenced by the number of scenarios included. Therefore, there is a need to enhance problem-solving techniques to reduce computation time and accelerate the solution process. Our study primarily concentrated on tactical and operational planning decisions, excluding the strategic decision-making related to the design of the supply chain network. This exclusion was because the case company under investigation already possesses its own supply chain network, comprising production units, cross docks, and external warehouses. Nevertheless, it could be an intriguing direction for future research to explore how providing customers with various strategies might influence changes in the design of the supply chain network, potentially leading to the opening, closing, or relocation of existing warehouses. Ultimately, suggesting a deeper exploration of operational specifics for each strategy could be a valuable direction for future research. Within this context, considering alternative

inventory control policies for VMI customers or extending lead times for MTO customer demands could be explored.

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Data availability statement

The data used in this study is available upon request. Interested researchers may contact the corresponding author to obtain access to the dataset and related materials. The data will be provided in a format that ensures confidentiality and compliance with applicable privacy regulations.

Disclosure statement

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Appendix

Table A1. Summary of the main characteristics of this study and of the reviewed literature on production-inventory systems.

References	Customer-wise	Considered strategy	Main decisions	Main considerations	Supply chain configuration	Number of products	Number of customers	Uncertainty	Objective	Methodology	Case study
Rajagopalan (2002)		MTS/MTO	Strategy assignment	Service level, lead time	S	m	–	*	Min	NLIP	Mold production
Huiskonen, Niemi, and Pirttilä (2003)	*	MTS/MTO	Customer characteristics			m	m			AF	
Småros et al. (2003)	*	VMI/non-VMI	Number of VMI customers, replenishment frequency, production planning length		T	s	m		Min	Sim	
Ghose et al. (2005)		MTS/MTO	Pricing strategy, production strategy	Fulfill demand	T	s	m		Max	GT	
Feng, D'Amours, and Beauregard (2008)		MTO	Sales quantity, production, distribution, and procurement planning	Backorder, limited capacity, shipping, production, inventory constraints	M	m	m		Max	MIP	Oriented Strand Board (OSB) manufacturing company
Kaminsky and Kaya (2009)		MTS/MTO	Production strategy, due date	Delivery, lead time, inventory, tardiness	T	m	s	*	Min	NLP	
Feng, D'Amours, and Beauregard (2010)		MTO	Sales quantity, production, distribution, and procurement planning	Sales, production, distribution, procurement constraints	M	m	m	*	Max	Sim	Oriented Strand Board (OSB) manufacturing company
Köber and Heinecke (2012)		MTS/MTO	Production strategy	Production time, customer desired delivery time	S	m	m			SD	Agricultural machinery
Zhang et al. (2013)	*	MTS/MTO	Capacity, inventory control parameters	FIFO rule for MTO, order-up-to for MTS	S	s	–	*	Min	QS	
Carlsson, Flisberg, and Rönnqvist (2014)		VMI	Transportation, inventory	Transport limitation, safety stock	M	m	m	*	Min	LP	Pulp producer
Lin and Wu (2014)			Product price, lost sales, production, shipping	Price-dependent stochastic demand	S	m		*	Max		
Choi and Cheol (2014)	*	MTS/MTO	Sales price, produced quantity, demand	Limited capacity, lost sales, backorder, price dependent demand	S	s	–		Max	NLP	
Adeinat and Ventura (2015)			Selling price and replenishment policy	Quantity discounts, price-sensitive demand	T	s	m		Max	MINLP	
Borade and Sweeney (2015)		VMI	The number of retailers to be served	Perishable product, service level	T	m	s		Min	GA	
Taleizadeh, Noori-Daryan, and Cárdenas-Barrón (2015)	*	VMI	Order quantity, selling price	Price-sensitive demand	M	s	s		Max	SG	
Liu, Shang, and Su (2016)			Sales effort level, pricing, and inventory	Price-sensitive demand	T	s	s	*	Max	NLP	
Jadidi, Jaber, and Zolfaghari (2017)			Pricing and inventory decisions	Price-sensitive demand, short life products, quantity discounts, limited capacity	T	s	m	*	Max	NLP	

(continued).

Table A1. Continued.

References	Customer-wise	Considered strategy	Main decisions	Main considerations	Supply chain configuration	Number of products	Number of customers	Uncertainty	Objective	Methodology	Case study
Ali et al. (2019)		MTS	Sales and operational planning	limited capacity, seasonal market, segmented customers	T	m	m		Max	Sim	Lumber industry
Kuthambalayan and Bera (2020b)		MTS/MTO	Production strategy, inventory, production planning length	Limited capacity, lost sales, selling price as parameter	T	m	–	*	Max	NLP	
Wang, Diabat, and Wu (2021)			Order quantity and the retail price for each product	Price-sensitive demand, order quantity requirement	T	s	s	*	Max	NLP	
Weißhuhn and Hoberg (2021)	*	VMI/non-VMI	Customer reorder point, replenishment quantity	Lost sales, capacity, buyer division	T	s	m	*	Max	NLP	
Duan and Ventura (2021)			Pricing, supplier selection, and inventory replenishment	Price-sensitive demand	T	s	s		Max	NLP	
Adeinat et al. (2022)			Selling price, order quantity	Price-sensitive demand	M	s	m		Max	MINLP, SA	
This study		MTS/MTO/VMI	Strategy assignment, selling price, operational decisions	Capacity, service level, backorder, price sensitive demand	M	m	m	*	Max	Two-stage stochastic MINLP	Pulp and paper industry

Supply chain configuration: S: single-stage facility; T: two-level; M: multi-level. **Number of products:** s: single; m: multiple. **Number of customers:** s: single; m: multiple. **Objective:** Min: minimisation of cost; Max: maximisation of profit. **Methodology:** NLIP: nonlinear integer programming; NLP: nonlinear programming; NL: nonlinear programming; MIP: mixed integer programming; MINLP: mixed integer nonlinear programming. GA: genetic algorithm; SA: simulated annealing; GT: game theory; AF: analytical framework; SD: System Dynamics; Sim: simulation; QS: queuing system; SG: Stackelberg game.